INCH-POUND

MIL-M-28787/351
26 October 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

68000 SUPPORT MODULE, KEY CODE ASC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic 68000 support module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.5995 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be ASC.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end of life	II	Q	1	INF	1	INF	1	INF	MOhm
A2-A16	Isolation, initial	II	X	10	INF	10	INF	10	INF	MOhm
	end of life	II	Q	1	INF	1	INF	1	INF	MOhm
B1-B4	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B5-B8	Continuity	III	Q	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C	Vcc stress	IV	Q	--------Module shall pass all remaining tests----------						
D1-D22	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D23-D33	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D34, D35	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D36	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D37, D38	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	mA
D39-D42	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D43, D44	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D45	Ibd	V	S	0.0	4.0	0.0	4.0	0.0	4.0	mA
D46	Ibd	V	S	0.0	1.4	0.0	1.4	0.0	1.4	mA
D47-D49	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D50	Ibd	V	S	0.0	8.0	0.0	8.0	0.0	8.0	mA
D51	Ibd	V	S	0.0	3.0	0.0	3.0	0.0	3.0	mA
E1-E31	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
E32, E33	Vic	VI	S	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	V
E34-E39	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
E40-E46	Vic	VI	S	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	V
E47-E49	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
E50, E51	Vic	VI	S	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
F1-F5	Ios	VII	S	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	mA
F6-F9	Ios	VII	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
F10	Ios	VII	S	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	mA
F11-F14	Ios	VII	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
F15,F16	Ios	VII	S	-110.0	-40.0	-110.0	-40.0	-110.0	-40.0	mA
F17	Ios	VII	S	-111.6	-40.7	-111.6	-40.7	-111.6	-40.7	mA
F18	Ios	VII	S	-101.6	-40.7	-101.6	-40.7	-101.6	-40.7	mA
F19,F20	Ios	VII	S	-112.2	-41.0	-112.2	-41.0	-112.2	-41.0	mA
F21	Ios	VII	S	-112.0	-41.0	-112.0	-41.0	-112.0	-41.0	mA
F22	Ios	VII	S	-104.0	-42.0	-104.0	-42.0	-104.0	-42.0	mA
F23	Ios	VII	S	-114.2	-42.0	-114.2	-42.0	-114.2	-42.0	mA
F24	Ios	VII	S	-113.4	-43.6	-113.4	-43.6	-113.4	-43.6	mA
F25	Ios	VII	S	-114.0	-42.0	-114.0	-42.0	-114.0	-42.0	mA
F26,F27	Ios	VII	S	-132.0	-31.0	-132.0	-31.0	-132.0	-31.0	mA
G1-G21	Iih	VIII	X	0.0	20.0	0.0	20.0	0.0	20.0	uA
G22	Iih	VIII	X	-295.0	-247.0	-295.0	-247.0	-295.0	-247.0	uA
G23-G31	Iih	VIII	X	0.0	25.0	0.0	25.0	0.0	25.0	uA
G32	Iih	VIII	X	0.0	100.0	0.0	100.0	0.0	100.0	uA
G33	Iih	VIII	X	0.0	50.0	0.0	50.0	0.0	50.0	uA
G34	Iih	VIII	X	-295.0	-207.0	-295.0	-207.0	-295.0	-207.0	uA
G35	Iih	VIII	X	0.0	80.0	0.0	60.0	0.0	60.0	uA
G36	Iih	VIII	X	0.0	80.0	0.0	80.0	0.0	80.0	uA
G37,G38	Iih	VIII	X	0.0	65.0	0.0	65.0	0.0	65.0	uA
G39	Iih	VIII	X	0.0	45.0	0.0	45.0	0.0	45.0	uA
G40-G42	Iih	VIII	X	0.0	70.0	0.0	70.0	0.0	70.0	uA
G43	Iih	VIII	X	0.0	100.0	0.0	100.0	0.0	100.0	uA
G44	Iih	VIII	X	0.0	200.0	0.0	200.0	0.0	200.0	uA
G45	Iih	VIII	X	0.0	400.0	0.0	400.0	0.0	400.0	uA
G46	Iih	VIII	X	0.0	100.0	0.0	100.0	0.0	100.0	uA
G47-G49	Iih	VIII	X	0.0	40.0	0.0	40.0	0.0	40.0	mA
G50	Iih	VIII	X	-2.95	-1.92	-2.95	-1.92	-2.95	-1.92	mA
G51	Iih	VIII	X	-2.95	-2.22	-2.95	-2.22	-2.95	-2.22	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
H1-H9	Iil	IX	X	-440.0	0.0	-440.0	0.0	-440.0	0.0	uA
H10-H16	Iil	IX	X	-400.0	0.0	-400.0	0.0	-400.0	0.0	uA
H17-H21	Iil	IX	X	-400.0	-30.0	-400.0	-30.0	-400.0	-30.0	uA
H22	Iil	IX	X	-937.0	-516.0	-937 0	-516.0	-937.0	-516.0	uA
H23-H31	Iil	IX	X	-250.0	0.0	-250.0	0.0	-250.0	0.0	uA
H32	Iil	IX	X	-760.0	-60.0	-760 0	-60.0	-760.0	-60.0	uA
H33	Iil	IX	X	-650.0	0 0	-650 0	0 0	-650.0	0.0	uA
H34	Iil	IX	X	-800.0	-30.0	-800.0	-30.0	-800.0	-30.0	uA
H35	Iil	IX	X	-630.0	0.0	-630.0	0.0	-630.0	0.0	uA
H36	Iil	IX	X	-880.0	0.0	-880.0	0.0	-880.0	0.0	uA
H37	Iil	IX	X	-1.34	-0.55	-1.34	-0.55	-1.34	-0.55	mA
H38	Iil	IX	X	-1.26	-0.06	-1.26	-0 06	-1.26	-0 06	mA
H39,H40	Iil	IX	X	-1.05	-0.03	-1 05	-0.03	-1 05	-0.03	mA
H41-H43	Iil	IX	X	-2.44	-1 06	-2 44	-1 06	-2 44	-1.06	mA
H44	Iil	IX	X	-4.76	-0.5	-4.76	-0.5	-4.76	-0.5	mA
H45	Iil	IX	X	-13.7	-5.72	-13.7	-5.72	-13.7	-5.72	mA
H46	Iil	IX	X	-51.12	-6.78	-51 12	-6 78	-51 12	-6 78	mA
H47	Iil	IX	X	-4.0	-0.5	-4.0	-0.5	-4.0	-0.5	mA
H48	Iil	IX	X	-2.0	-1.0	-2.0	-1 0	-2.0	-1.0	mA
H49	Iil	IX	X	-3.2	-0.5	-3.2	-0.5	-3 2	-0.5	mA
H50	Iil	IX	X	-8.0	-0.5	-8.0	-0.5	-8 0	-0.5	mA
H51	Iil	IX	X	-16.0	-0.5	-16 0	-0.5	-16 0	-0 5	mA
J1-J11	Voh	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
J12-J17	Voh	X	X	2.4	4.5	2.4	4 5	2.4	4.5	V
J18-J27	Voh	X	X	2.5	4.5	2.5	4 5	2.5	4.5	V
J28	Voh	X	X	2.4	2.6	2.4	2 6	2.4	2.6	V
K1-K10	Vol	XI	X	0.0	0.4	0.0	0 4	0.0	0 4	V
K11-K20	Vol	XI	X	0.0	0.5	0.0	0 5	0.0	0.5	V
K21-K27	Vol	XI	X	0.2	0.5	0.2	0.5	0.2	0.5	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
L1-L4	Iozh	XII	X	-20	20	-20	20	-20	20	uA
M1-M4	Iozl	XIII	X	-20	20	-20	20	-20	20	uA
N	Icc	XIV	X	30.4	609.0	30 4	609.0	30.4	609.0	mA
P1,P2	Functional	XV	X	----------See table XV----------						
Q1,Q2	Dynamic	XVI	S	----------See table XVI----------						
Q3,Q4	Dynamic	XVI	Q	----------See table XVI----------						
R	PSFC	XVIII	S	4.0	7.6	4.0	7.6	4.0	7.6	uF
S	PSTR	XIX	Q	0	400	0	400	0	400	mV
T1-T11	Output crosstalk (high level)	XX	Q	0	400	0	400	0	400	mV
T12-T28	Output crosstalk (high level)	XX	Q	0	500	0	500	0	500	mV
U1-U6	Output crosstalk (low level)	XXI	Q	0	400	0	400	0	400	mV
U7-U23	Output crosstalk (low level)	XXI	Q	0	500	0	500	0	500	mV
U24-U27	Output crosstalk (low level)	XXI	Q	0	650	0	650	0	650	mV

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
V1-V4	Crosstalk duration (low level)	XXII	Q	0	50	0	50	0	50	nS
W1-W19	Cin	XXIII	Q	2	20	2	20	2	20	pF
W20,W21	Cin	XXIII	Q	2	25	2	25	2	25	pF
W22-W28	Cin	XXIII	Q	2	15	2	15	2	15	pF
W29,W30	Cin	XXIII	Q	4	30	4	30	4	30	pF
W31-W36	Cin	XXIII	Q	2	40	2	40	2	40	pF
W37-W39	Cin	XXIII	Q	4	40	4	40	4	40	pF
W40-W43	Cin	XXIII	Q	2	50	2	50	2	50	pF
W44	Cin	XXIII	Q	4	60	4	60	4	60	pF
W45,W46	Cin	XXIII	Q	4	45	4	45	4	45	pF
W47,W48	Cin	XXIII	Q	6	45	6	45	6	45	pF
W49	Cin	XXIII	Q	8	75	8	75	8	75	pF
X1-X4	Cout	XXIV	Q	2	25	2	25	2	25	pF
Y1-Y4	Disable time	XXV	Q	3	31	3	31	3	31	nS
Y5-Y8	Disable time	XXV	Q	3	69	3	69	3	69	nS
Z	Threshold	XXVI	X	See table XXVI						

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/351-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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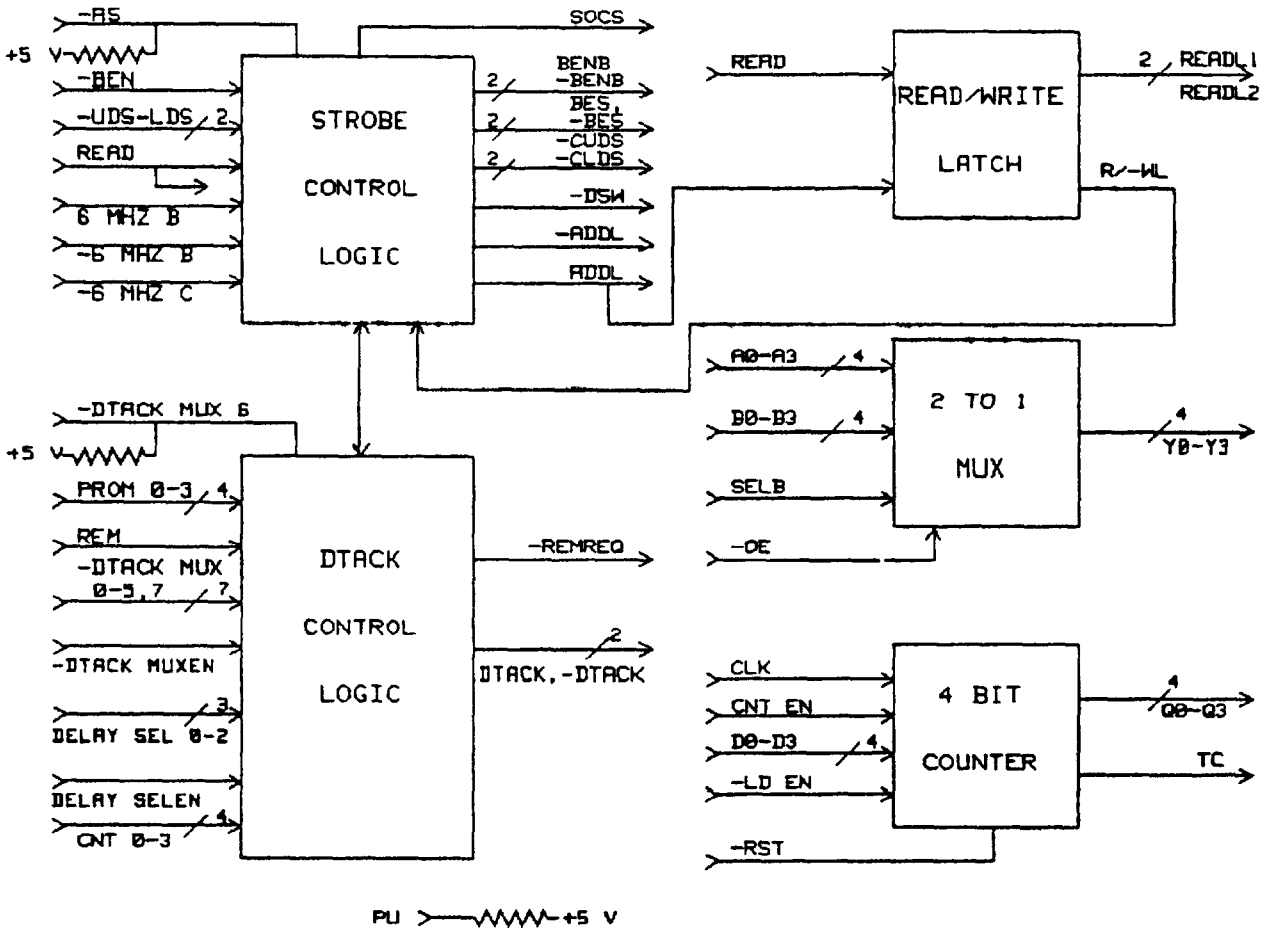


FIGURE 1. ASC functional block diagram.

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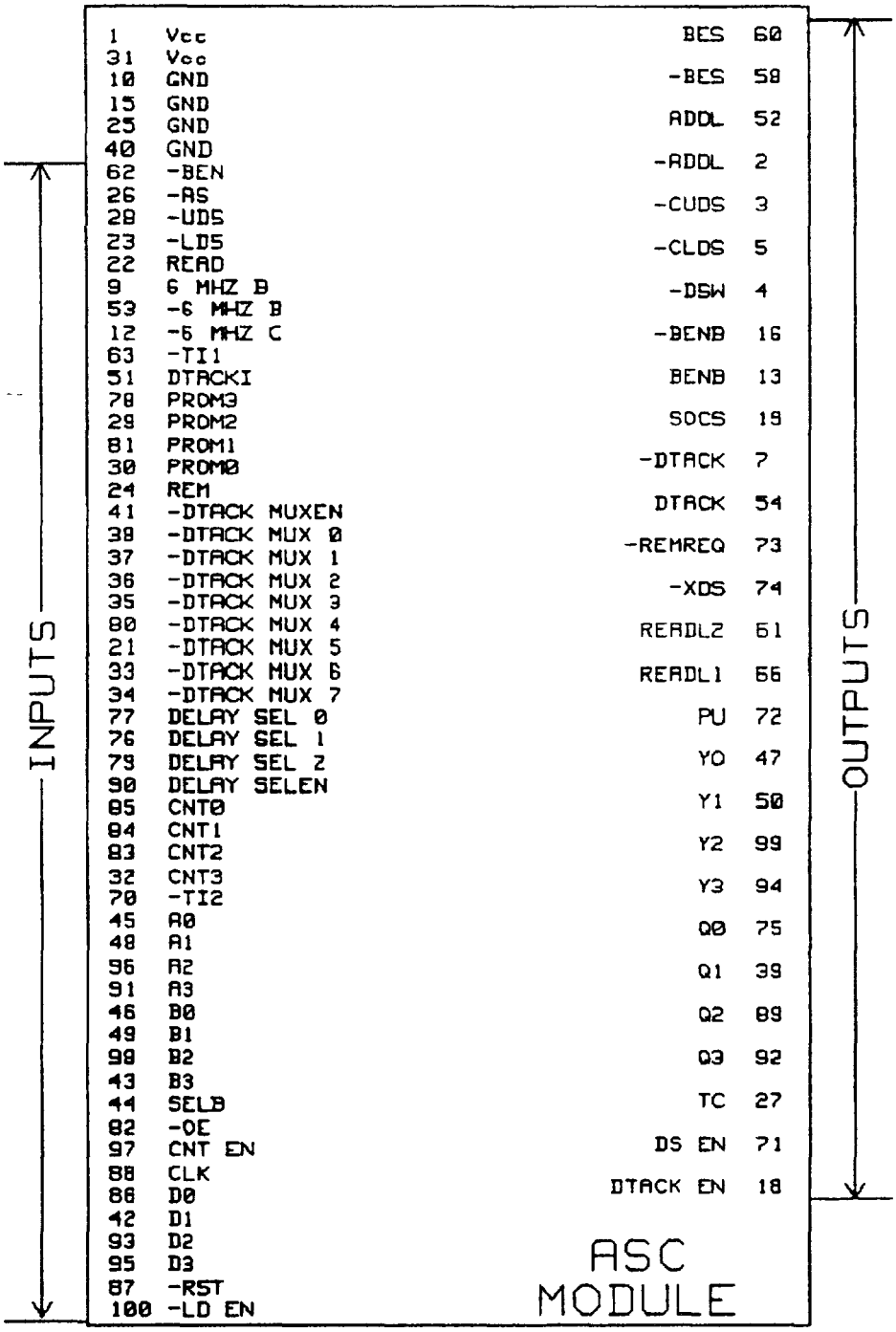


FIGURE 2. ASC pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and F where applicable, followed by the remaining tests in the sequence.

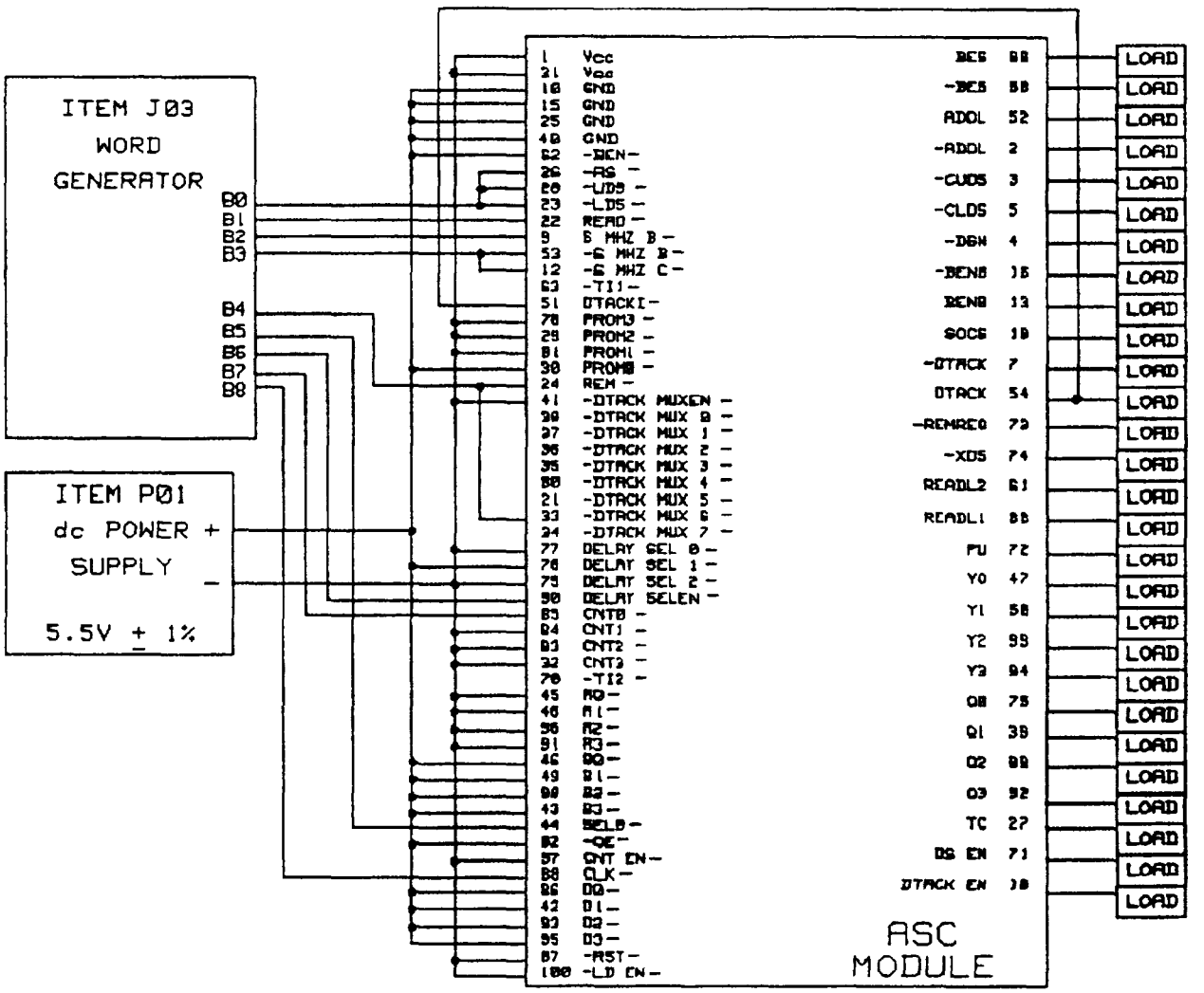
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through V, Y and Z shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTES:

1. Load each output as specified in the output voltage test.

FIGURE 3. ASC operating temperature and life test circuit.

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2. The word generator vectors are as follows.

WORD	B0	B1	B2	B3	B4	B5	B6	B7	B8
1	1	1	1	0	0	0	0	0	0
2	1	1	1	0	0	0	0	0	1
3	1	1	0	1	0	0	0	0	0
4	1	0	0	1	0	0	0	0	1
5	0	0	1	0	0	1	0	0	0
6	0	0	1	0	0	1	0	0	1
7	0	0	0	1	0	1	0	0	0
8	0	0	0	1	0	1	1	1	1
9	0	0	1	0	0	0	1	1	0
10	0	0	1	0	1	0	1	1	1
11	0	0	0	1	1	0	1	1	0
12	0	0	0	1	1	0	1	1	1
13	0	0	1	0	1	1	1	1	0
14	0	0	1	0	1	1	1	1	1
15	0	1	0	1	1	1	0	1	0
16	1	1	0	1	1	1	0	1	1

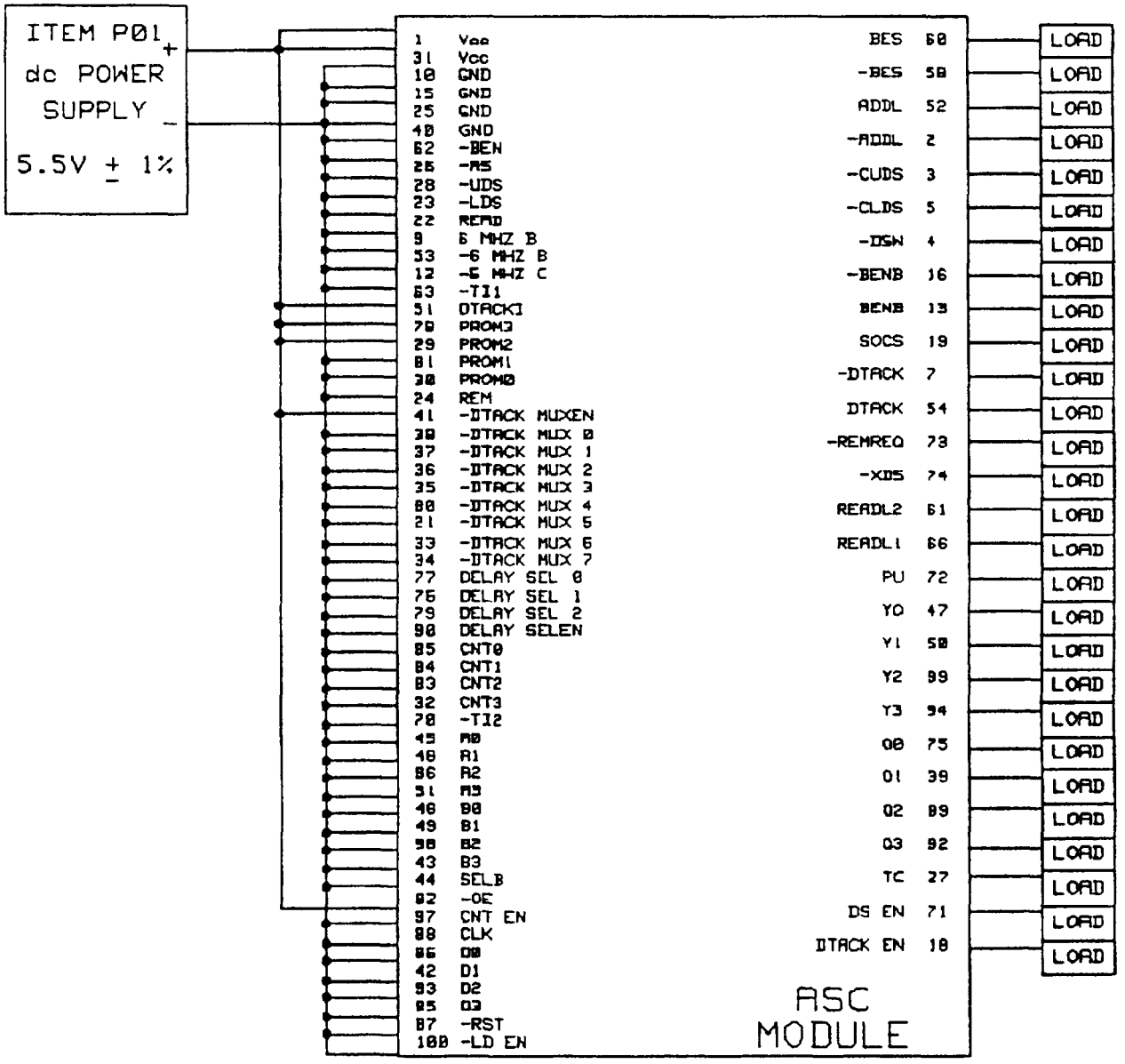
3. Rise and fall times of inputs shall not exceed 30 nS.

4. Word duration shall be 1 uS $\pm$ 100 nS.

5. Logic 1 = 3.0 $\pm$ 100 mV.
   Logic 0 = 0.0 $\pm$ 100 mV.

FIGURE 3. ASC operating temperature and life test circuit - Continued.

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NOTE:

1. Load each output as specified in the output voltage test.

FIGURE 4. ASC component temperature test circuit.

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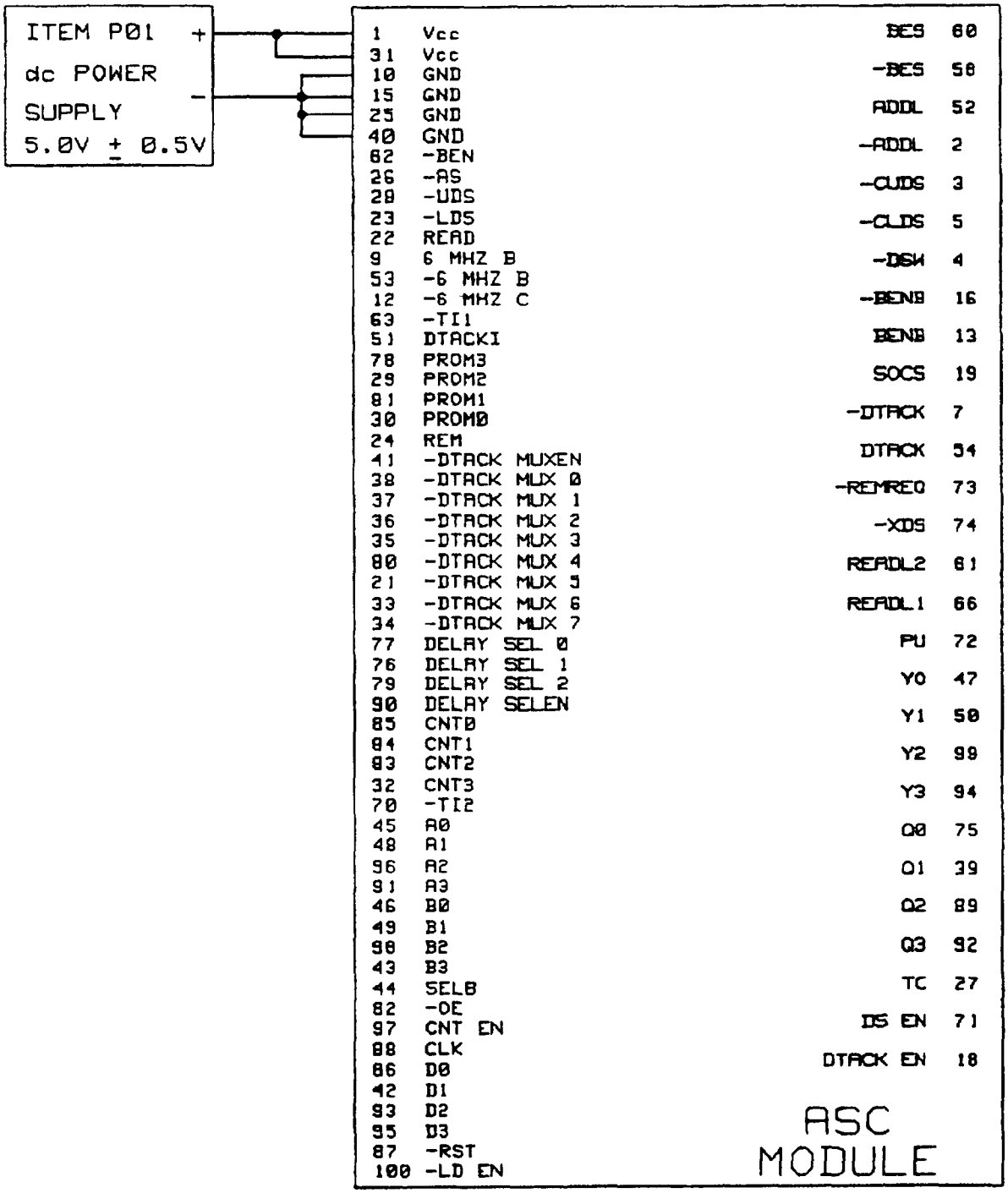


FIGURE 5. ASC vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests B, G through V, Y and Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through V, Y and Z shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 3.349 W (max); 1.496 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (mA)	Cin max (pF)
A0	-1.5	0.7	2.0	-0.44	20	0.1	20
A1	-1.5	0.7	2.0	-0.44	20	0.1	20
A2	-1.5	0.7	2.0	-0.44	20	0.1	20
A3	-1.5	0.7	2.0	-0.44	20	0.1	20
B0	-1.5	0.7	2.0	-0.44	20	0.1	20
B1	-1.5	0.7	2.0	-0.44	20	0.1	20
B2	-1.5	0.7	2.0	-0.44	20	0.1	20
B3	-1.5	0.7	2.0	-0.44	20	0.1	20
-OE	-1.5	0.7	2.0	-0.44	20	0.1	25
SELB	-1.5	0.7	2.0	-0.88	40	0.2	20
D0	-1.5	0.7	2.0	-0.40	20	0.1	25
D1	-1.5	0.7	2.0	-0.40	20	0.1	25
D2	-1.5	0.7	2.0	-0.40	20	0.1	25
D3	-1.5	0.7	2.0	-0.40	20	0.1	25

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (mA)	Cin max (pF)
CLK	-1.5	0.7	2.0	-0.63	40	0.2	20
CNT EN	-1.5	0.7	2.0	-1.26	60	0.3	30
-RST	-1.5	0.7	2.0	-0.40	20	0.1	20
-LD EN	-1.5	0.7	2.0	-0.80	40	0.2	20
-DTACK MUX 0	-1.5	0.7	2.0	-0.40	20	0.1	15
-DTACK MUX 1	-1.5	0.7	2.0	-0.40	20	0.1	15
-DTACK MUX 2	-1.5	0.7	2.0	-0.40	20	0.1	15
-DTACK MUX 3	-1.5	0.7	2.0	-0.40	20	0.1	15
-DTACK MUX 4	-1.5	0.7	2.0	-0.40	20	0.1	15
-DTACK MUX 7	-1.5	0.7	2.0	-0.40	20	0.1	15
-DTACK MUX 6	-1.5	0.7	2.0	-1.34	-295	0.3	20
-DTACK MUX 5	-1.5	0.7	2.0	-0.76	80	0.4	40
PROM1	-1.5	0.7	2.0	-1.05	65	1.2	45
PROM2	-1.5	0.7	2.0	-1.05	65	1.2	45

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (mA)	Cin max (pF)
PROM0	-1.5	0.7	2.0	-0.65	45	1.1	40
-DTACK MUX EN	-1.5	0.7	2.0	-0.40	20	0.1	15
-AS	-1.5	0.7	2.0	-0.94	-295	0.1	20
READ	-1.2	0.7	2.0	-2.44	70	1.1	40
-UDS	-1.2	0.7	2.0	-2.44	70	1.1	40
-LDS	-1.2	0.7	2.0	-2.44	70	1.1	40
CNT0	-1.5	0.8	2.0	-0.25	25	1.0	50
CNT1	-1.5	0.8	2.0	-0.25	25	1.0	50
CNT2	-1.5	0.8	2.0	-0.25	25	1.0	50
CNT3	-1.5	0.8	2.0	-0.25	25	1.0	50
DELAY SEL 0	-1.5	0.8	2.0	-0.25	25	1.0	50
DELAY SEL 1	-1.5	0.8	2.0	-0.25	25	1.0	50
DELAY SEL 2	-1.5	0.8	2.0	-0.25	25	1.0	50
DELAY SEL EN	-1.5	0.8	2.0	-0.25	25	1.0	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (mA)	Cin max (pF)
PROM 3	-1.5	0.8	2.0	-0.25	25	1.0	50
-BEN	-1.2	0.8	2.0	-2.0	50	1.0	20
DTACKI	-1.2	0.8	2.0	-3.2	100	2.0	45
6 MHZ B	-1.2	0.8	2.0	-8.0	200	2.0	60
-6 MHZ B	-1.2	0.8	2.0	-16.0	400	4.0	75
-6 MHZ C	-1.2	0.8	2.0	-4.76	180	1.4	45
REM	-1.2	0.8	2.0	-4.0	100	1.0	20

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh min (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
Y0	2.4	0.4	-1.0	12.0	20	-20	-15.0	25
Y1	2.4	0.4	-1.0	12.0	20	-20	-15.0	25
Y2	2.4	0.4	-1.0	12.0	20	-20	-15.0	25
Y3	2.4	0.4	-1.0	12.0	20	-20	-15.0	25
Q0	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
Q1	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
Q2	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
Q3	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
TC	2.5	0.4	-0.4	4.0	NA	NA	-15.0	NA
-XDS	2.5	0.4	-0.38	3.8	NA	NA	-15.0	NA
-CUDS	2.5	0.5	-1.0	20.0	NA	NA	-40.0	NA
-CLDS	2.5	0.5	-1.0	20.0	NA	NA	-40.0	NA
READL1	2.5	0.5	-1.0	20.0	NA	NA	-40.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
READL2	2.5	0.5	-1.0	20.0	NA	NA	-40.0	NA
DTACK	2.5	0.5	-1.0	20.0	NA	NA	-40.0	NA
-ADDL	2.5	0.5	-0.95	18.4	NA	NA	-40.7	NA
ADDL	2.5	0.5	-0.87	15.8	NA	NA	-42.0	NA
-DSW	2.5	0.5	-0.95	18.0	NA	NA	-41.0	NA
BENB	2.5	0.5	-0.90	16.0	NA	NA	-42.0	NA
-BENB	2.5	0.5	-0.85	14.0	NA	NA	-42.0	NA
-DTACK	2.5	0.5	-0.95	18.0	NA	NA	-41.0	NA
SOCS	2.5	0.5	-0.69	7.0	NA	NA	-43.6	NA
-BES	2.5	0.5	-0.95	18.4	NA	NA	-40.7	NA
BES	2.5	0.5	-0.93	17.75	NA	NA	-41.0	NA
-REMREQ	2.5	0.5	-1.0	20.0	NA	NA	-40.0	NA
DS EN	2.4	0.5	-2.0	8.0	NA	NA	-31.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
DTACK EN	2.4	0.5	-2.0	8.0	NA	NA	-31.0	NA
PU	2.4	NA	-2.0	NA	NA	NA	NA	NA

6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The module function is printed herein in appendix.

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6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
A0	45	I	These are four inputs to the quad 2-input multiplexer. When SELB is low, A0-A3 correspond to Y0-Y3 (respectively).
A1	48	I	
A2	96	I	
A3	91	I	
B0	46	I	Four inputs to the quad 2-input multiplexer. When SELB is high, B0-B3 correspond to Y0-Y3 (respectively).
B1	49	I	
B2	98	I	
B3	43	I	
-OE	82	I	This is the active low output enable signal for the quad 2-input multiplexer. When -OE is low, Y0-Y3 are enabled. When -OE is high, Y0-Y3 are in the high impedance state.
-DTACK MUX 0	38	I	These are the active low inputs for peripherals that have a data transfer acknowledge output. The proper DTACK MUX input is selected by placing the proper binary number on PROM0 (LSB), PROM1, and PROM2 (MSB).
-DTACK MUX 1	37	I	
-DTACK MUX 2	36	I	
-DTACK MUX 3	35	I	
-DTACK MUX 4	80	I	
-DTACK MUX 5	21	I	
-DTACK MUX 6	33	I	
-DTACK MUX 7	34	I	
-DTACK MUXEN	41	I	An active low input that enables the DTACK multiplexer. If this pin is high and PROM3 is low, DTACK EN will be generated without concern for other DTACK MUX inputs.
D0	86	I	These input signals are the preset inputs to the 4-bit binary counter. Presetting is synchronous, which means setting up a low level at the counter load input (-LD EN) disables the counter sequence and causes the counter outputs to agree with the preset inputs, after the next clock pulse, regardless of the level of the counter enable (CNT EN) input.
D1	42	I	
D2	93	I	
D3	95	I	

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-RST	87	I	This active low input signal provides an asynchronous clear of all four counter outputs regardless of the levels of other inputs.
-AS	26	I	An active low input which indicates that there is a valid address on the address bus. This input is internally pulled up.
DELAY SEL 0 DELAY SEL 1 DELAY SEL 2	77 76 79	I I I	These inputs define the type of peripheral being accessed during a bus cycle for use in generating data strobes (DS EN, -CUDS, -CLDS, and -DSW) and data transfer acknowledge (DTACK EN, -DTACK and DTACK).
DELAY SELEN	90	I	This active high input signal is used in generating data strobes (DS EN, -DSW, -CUDS, and -CLDS).
CNT0 CNT1 CNT2 CNT3	85 84 83 32	I I I I	These input signals provide control over the timing of DTACK EN.
PROM0 PROM1 PROM2 PROM3	30 81 29 78	I I I I	These inputs define the type of peripheral being accessed during a bus cycle for use in generating data strobes (DS EN, -DSW, -CUDS and -CLDS) and data transfer acknowledge (DTACK EN, -DTACK and DTACK). PROM1 and PROM2 are also used to generate -REMREQ.
REM	24	I	This rising edge sensitive signal clocks an internal flip-flop, which generates -REMREQ.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-BEN	62	I	An active low input which indicates that the 68000 is supervising transfers. When Direct Memory Access (DMA) or another supervisor is controlling transfers, -BEN should be high.
CNT EN	97	I	This active high input signal enables the 4 bit binary counter for counting. When CNT EN is low, the count sequence is disabled.
-UDS	28	I	An active low input which indicates that there are valid data on bits 8-15 of the data bus.
-LDS	23	I	An active low input which indicates that there are valid data on bits 0-7 of the data bus.
READ	22	I	This input defines the data bus transfer as a supervisor read (high) or supervisor write (low).
DTACKI	51	I	This input shall be tied to DTACK.
6 MHZ B -6 MHZ B -6 MHZ C	9 53 12	I I I	These clock inputs are used by the ASC strobe control logic to generate and synchronize control signals. 6 MHZ B should be generated by buffering the 68000 clock. -6 MHZ B and -6 MHZ C should be complements of 6 MHZ B.
-LD EN	100	I	This input signal, when active (low), enables the preset inputs (D0-D3) to be loaded into the 4-bit counter on the next clock pulse independent of the counter enable (CNT EN) input.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
CLK	88	I	This is the clock input for the synchronous 4-bit binary counter. The counter outputs (Q0-Q3) change state on the rising edge of the clock input when the counter is enabled (CNT EN is high) or during a parallel load (-LD EN is low).
SELB	44	I	This is the quad 2-input multiplexer select input. When SELB is low, A0-A3 are selected as inputs. When SELB is high, B0-B3 are selected as inputs.
-TI1 -TI2	63 70	I I	These two test inputs perform no user function and should be left open.
Q0 Q1 Q2 Q3	75 39 89 92	O O O O	These signals are the outputs of the 4-bit counter. In the Trident II Fire Control System application, these outputs are connected to the ASC module counter inputs (CNT0-CNT3).
TC	27	O	This output signal is the terminal count output of the 4-bit counter. It is active (high) when the counter has achieved terminal count (all 1's) and the count enable (CNT EN) input is active (high).
-CUDS	3	O	This active low output signal is a function of -UDS, -AS, and DS EN. The signal becomes active one clock cycle after DS EN and goes inactive when -AS or -UDS are negated.
-CLDS	5	O	This active low output signal is a function of -LDS, -AS and DS EN. The signal becomes active one clock cycle after DS EN and goes inactive when -AS or -LDS are negated.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
READL1 READL2	66 61	O O	These outputs are generated by a transparent latch. When ADDL is low, READL1 and READL2 correspond to READ. If READ is low (write) when ADDL goes active, READL1 and READL2 will be latched low until ADDL is negated.
-XDS	74	O	This active low output signal is the output of the external DTACK multiplexer synchronized to the falling edge of -6 MHZ C.
Y0 Y1 Y2 Y3	47 50 99 94	O O O O	These are the four tri-stateable outputs of the quad 2-input multiplexer.
-REMREQ	73	O	This active low output signal is active after the rising edge of REM when PROM1 and PROM2 are both high. The output becomes inactive when -AS or -UDS, -LDS and READ are negated.
DTACK	54	O	This active high output indicates that a data transfer is complete. DTACK becomes active on the falling edge of -6 MHZ B following DTACK EN and becomes inactive when -AS or -UDS, -LDS and READ are negated.
-ADDL	2	O	This active low output signal is the logical complement of ADDL.
-BES	58	O	This active low output is the complement of BES.
BES	60	O	This active high output indicates that the address and data busses contain valid data.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-DTACK	7	O	This active low output is the logical complement of DTACK.
-DSW	4	O	This active low output signal is a function of DS EN, -AS and READ. -DSW becomes active one clock cycle after DS EN and goes inactive when -AS or READ go high.
BENB	13	O	An active high output which is generated by inverting -BEN.
ADDL	52	O	An active high output signal which becomes active one clock cycle after DTACK and is negated on the falling edge of -6 MHZ B after BES becomes inactive.
SOCS	19	O	This active high output signal signifies the start of a bus cycle.
-BENB	16	O	An active low output that is generated by buffering -BEN.
DS EN	71	O	This active high output signal is used internally to enable the data strobe generation logic. This pin is primarily intended as a test point.
DTACK EN	18	O	This active high output signal is used internally to enable the DTACK generation logic. It is primarily pinned out as a test point and has little functional value.
PU	72	O	This is an internal connection to a 1 KOhm pull-up resistor. It is available for general purpose use in tying module inputs to a logic high.

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 6.

(j) Vol test - table XI, figure 6.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

(n) Functional test - table XV, figure 7.

(o) Dynamic test - tables XVI and XVII, figures 8 and 9.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figure 10.

(r) Output crosstalk tests - tables XX and XXI, figures 11 and 12.

(s) Output crosstalk duration test - table XXII, figure 13.

(t) Cin test - table XXIII.

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(u) Cout test - table XXIV.

(v) Disable time test - table XXV, figures 14 and 15.

(w) Threshold test - table XXVI.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT, AV
Navy - AS, MC
Air Force - 13, 17, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0051)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	All pins	Frame	10	INF	MOhm	1-7
A2	Isolation, used to unused pins	All used pins	All unused pins	10	INF	MOhm	1-7
A3	Isolation, unused to unused pins	6	8,11,14, 17,20,55, 56,57,59, 64,65,67, 68,69	10	INF	MOhm	1-7
A4	Isolation, unused to unused pins	8	11,14,17, 20,55,56, 57,59,64, 65,67,68, 69				
A5	Isolation, unused to unused pins	11	14,17,20, 55,56,57 59,64,65, 67,68,69				
A6	Isolation, unused to unused pins	14	17,20,55, 56,57,59, 64,65,67, 68,69				
A7	Isolation, unused to unused pins	17	20,55,56, 57,59,64, 65,67,68, 69				
A8	Isolation, unused to unused pins	20	55,56,57, 59,64,65, 67,68,69				
A9	Isolation, unused to unused pins	55	56,57,59, 64,65,67, 68,69	10	INF	Mohm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A10	Isolation, unused to unused pins	56	57,59,64, 65,67,68, 69	10	INF	MOhm	1-7
A11	Isolation, unused to unused pins	57	59,64,65, 67,68,69				
A12	Isolation, unused to unused pins	59	64,65,67, 68,69				
A13	Isolation, unused to unused pins	64	65,67,68, 69				
A14	Isolation, unused to unused pins	65	67,68,69				
A15	Isolation, unused to unused pins	67	68,69				
A16	Isolation, unused to unused pins	68	69	10	INF	MOhm	1-7

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V 30 V	± 10 percent ± 10 percent	A1-A2 A3-A16

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A1-A16 A1-A16

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, Vcc	1	31	0.0	1.0	Ohm	1-4
B2	Continuity, GND	10	15				
B3	Continuity, GND	15	25				
B4	Continuity, GND	25	40	0.0	1.0	Ohm	1-4
B5	Continuity, Vcc	1	31	0.0	0.1	Ohm	1-4
B6	Continuity, GND	10	15				
B7	Continuity, GND	15	25				
B8	Continuity, GND	25	40	0.0	0.1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 mOhm ± 5 mOhm	B1-B4 B5-B8

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31	10,15 25,40	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	914 mA	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	A0	45	7.0	0.0	0.1	mA	1-3
D2	A1	48					
D3	A2	96					
D4	A3	91					
D5	B0	46					
D6	B1	49					
D7	B2	98					
D8	B3	43					
D9	-OE	82					
D10	-DTACK MUX 0	38					
D11	-DTACK MUX 1	37					
D12	-DTACK MUX 2	36					
D13	-DTACK MUX 3	35					
D14	-DTACK MUX 4	80					
D15	-DTACK MUX 7	34					
D16	-DTACK MUXEN	41	7.0	0.0	0.1	mA	1-3
D17	D0	86	5.5	0.0	0.1	mA	1-3
D18	D1	42					
D19	D2	93					
D20	D3	95					
D21	-RST	87	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D22	-AS	26	5.5	0.0	0.1	mA	1-3
D23	DELAY SEL 0	77	5.5	0.0	1.0	mA	1-3
D24	DELAY SEL 1	76					
D25	DELAY SEL 2	79					
D26	DELAY SELEN	90					
D27	CNT0	85					
D28	CNT1	84					
D29	CNT2	83					
D30	CNT3	32					
D31	PROM3	78					
D32	REM	24					
D33	-BEN	62	5.5	0.0	1.0	mA	1-3
D34	-DTACK MUX 6	33	5.5	0.0	0.3	mA	1-3
D35	CNT EN	97	5.5	0.0	0.3	mA	1-3
D36	-DTACK MUX 5	21	5.5	0.0	0.4	mA	1-3
D37	PROM2	29	5.5	0.0	1.2	mA	1-3
D38	PROM1	81	5.5	0.0	1.2	mA	1-3
D39	PROM0	30	5.5	0.0	1.1	mA	1-3
D40	-UDS	28					
D41	-LDS	23					
D42	READ	22	5.5	0.0	1.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D43	DTACKI	51	5.5	0.0	2.0	mA	1-3
D44	6 MHZ B	9	5.5	0.0	2.0	mA	1-3
D45	-6 MHZ B	53	5.5	0.0	4.0	mA	1-3
D46	-6 MHZ C	12	5.5	0.0	1.4	mA	1-3
D47	-LD EN	100	5.5	0.0	0.2	mA	1-3
D48	CLK	88	5.5	0.0	0.2	mA	1-3
D49	SELB	44	7.0	0.0	0.2	mA	1-3
D50	-TI1	63	5.5	0.0	8.0	mA	1-3
D51	-TI2	70	5.5	0.0	3.0	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D21, D23-D33, D35-D51
	5.5 V	+0,-55 mV	D22,D34
Power supply current limit	0.913 A	NA	D1-D51
Forcing voltage, (Vin)	7.0 V	+0 mV,-77 mV	D1-D16
	5.5 V	± 55 mV	D17-D21
	5.5 V	-0 mV,+55 mV	D22
	5.5 V	± 55 mV	D23-D33

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing voltage, (Vin)	5.5 V	-0 mV,+55 mV	D34
	5.5 V	± 55 mV	D35-D48
	7.0 V	± 77 mV	D49
	5.5 V	± 55 mV	D50,D51
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D51
low, (VL)	0.0 V	± 100 mV	D1-D51
Input current measurement	NA	± 5.0 uA	D1-D22
	NA	± 50.0 uA	D23-D33
	NA	± 15.0 uA	D34,D35
	NA	± 20.0 uA	D36
	NA	± 60.0 uA	D37,D38
	NA	± 55.0 uA	D39-D42
	NA	± 100.0 uA	D43,D44
	NA	± 200.0 uA	D45
	NA	± 70.0 uA	D46
	NA	± 10.0 uA	D47-D49
	NA	± 0.4 mA	D50
	NA	± 150.0 uA	D51

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	I_{IN} (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E1	A0	45	-18.0	-1.5	-0.4	V	1-3
E2	A1	48					
E3	A2	96					
E4	A3	91					
E5	B0	46					
E6	B1	49					
E7	B2	98					
E8	B3	43					
E9	-OE	82					
E10	-DTACK MUX 0	38					
E11	-DTACK MUX 1	37					
E12	-DTACK MUX 2	36					
E13	-DTACK MUX 3	35					
E14	-DTACK MUX 4	80					
E15	-DTACK MUX 7	34					
E16	-DTACK MUXEN	41					
E17	D0	86					
E18	D1	42					
E19	D2	93					
E20	D3	95					
E21	-RST	87	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	-AS	26	-18.0	-1.5	-0.4	V	1-3
E23	DELAY SEL 0	77					
E24	DELAY SEL 1	76					
E25	DELAY SEL 2	79					
E26	DELAY SELEN	90					
E27	CNT0	85					
E28	CNT1	84					
E29	CNT2	83					
E30	CNT3	32					
E31	PROM3	78	-18.0	-1.5	-0.4	V	1-3
E32	REM	24	-18.0	-1.2	-0.4	V	1-3
E33	-BEN	62	-18.0	-1.2	-0.4	V	1-3
E34	-DTACK MUX 6	33	-18.0	-1.5	-0.4	V	1-3
E35	CNT EN	97					
E36	-DTACK MUX 5	21					
E37	PROM2	29					
E38	PROM1	81					
E39	PROM0	30	-18.0	-1.5	-0.4	V	1-3
E40	-UDS	28	-18.0	-1.2	-0.4	V	1-3
E41	-LDS	23					
E42	READ	22	-18.0	-1.2	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E43	DTACKI	51	-18.0	-1.2	-0.4	V	1-3
E44	6 MHZ B	9	"	"	"	"	1-3
E45	-6 MHZ B	53	"	"	"	"	1-4
E46	-6 MHZ C	12	-18.0	-1.2	-0.4	V	1-3
E47	-LD EN	100	-18.0	-1.5	-0.4	V	1-3
E48	CLK	88	"	"	"	"	"
E49	SELB	44	-18.0	-1.5	-0.4	V	1-3
E50	-TI1	63	-18.0	-1.2	-0.4	V	1-4
E51	-TI2	70	-18.0	-1.2	-0.4	V	1-4

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E51
Power supply current limit	0.913 A	NA	E1-E51
Forcing current, (Iin)	-18 mA	± 1.8 mA	E1-E51

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	E1-E51
low, (VL)	0.0 V	± 100 mV	E1-E51
Input voltage measurement	NA	± 55 mV	E1-E31
	NA	± 40 mV	E32,E33
	NA	± 55 mV	E34-E39
	NA	± 40 mV	E40-E46
	NA	± 55 mV	E47-E49
	NA	± 40 mV	E50,E51

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

4. Test conditions E45, E50 and E51 shall be performed with the power supply disconnected.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	Q0	75	0.0	-100.0	-15.0	mA	1-4
F2	Q1	39					
F3	Q2	89					
F4	Q3	92					
F5	TC	27	0.0	-100.0	-15.0	mA	1-4
F6	-CUDS	3	0.0	-100.0	-40.0	mA	1-4
F7	-CLDS	5					
F8	READL1	66					
F9	READL2	61	0.0	-100.0	-40.0	mA	1-4
F10	-XDS	74	0.0	-100.0	-15.0	mA	1-4
F11	Y0	47	0.0	-130.0	-15.0	mA	1-4
F12	Y1	50					
F13	Y2	99					
F14	Y3	94	0.0	-130.0	-15.0	mA	1-4
F15	-REMREQ	73	0.0	-110.0	-40.0	mA	1-4
F16	DTACK	54	0.0	-110.0	-40.0	mA	1-4
F17	-ADDL	2	0.0	-111.6	-40.7	mA	1-4
F18	-BES	58	0.0	-101.6	-40.7	mA	1-4
F19	BES	60	0.0	-112.2	-41.0	mA	1-4
F20	-DTACK	7	0.0	-112.2	-41.0	mA	1-4
F21	-DSW	4	0.0	-112.0	-41.0	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F22	BENB	13	0.0	-104.0	-42.0	mA	1-4
F23	ADDL	52	0.0	-114.2	-42.0	mA	1-4
F24	SOCS	19	0.0	-113.4	-43.6	mA	1-4
F25	-BENB	16	0.0	-114.0	-42.0	mA	1-4
F26	DS EN	71	0.0	-132.0	-31.0	mA	1-4
F27	DTACK EN	18	0.0	-132.0	-31.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F27
Power supply current limit	0.913 A	NA	F1-F27
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F27
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	F1-F27 F1-F27

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 4.25 mA	F1-F5
	NA	± 3.0 mA	F6-F9
	NA	± 4.25 mA	F10
	NA	± 5.75 mA	F11-F14
	NA	± 3.5 mA	F15,F16
	NA	± 3.54 mA	F17
	NA	± 3.04 mA	F18
	NA	± 3.56 mA	F19,F20
	NA	± 3.55 mA	F21
	NA	± 3.1 mA	F22
	NA	± 3.61 mA	F23
	NA	± 3.49 mA	F24
	NA	± 3.6 mA	F25
	NA	± 5.05 mA	F26,27

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

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NOTES: - Continued.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	A0	45	2.7	0.0	20.0	uA	1-3
G2	A1	48					
G3	A2	96					
G4	A3	91					
G5	B0	46					
G6	B1	49					
G7	B2	98					
G8	B3	43					
G9	-OE	82					
G10	-DTACK MUX 0	38					
G11	-DTACK MUX 1	37					
G12	-DTACK MUX 2	36					
G13	-DTACK MUX 3	35					
G14	-DTACK MUX 4	80					
G15	-DTACK MUX 7	34					
G16	-DTACK MUXEN	41					
G17	D0	86					
G18	D1	42					
G19	D2	93					
G20	D3	95					
G21	-RST	87	2.7	0.0	20.0	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G22	-AS	26	2.7	-295.0	-247.0	uA	1-3
G23	DELAY SEL 0	77	2.7	0.0	25.0	uA	1-3
G24	DELAY SEL 1	76					
G25	DELAY SEL 2	79					
G26	DELAY SELEN	90					
G27	CNT0	85					
G28	CNT1	84					
G29	CNT2	83					
G30	CNT3	32					
G31	PROM3	78	2.7	0.0	25.0	uA	1-3
G32	REM	24	2.7	0.0	100.0	uA	1-3
G33	-BEN	62	2.7	0.0	50.0	uA	1-3
G34	-DTACK MUX 6	33	2.7	-295.0	-207.0	uA	1-3
G35	CNT EN	97	2.7	0.0	60.0	uA	1-3
G36	-DTACK MUX 5	21	2.7	0.0	80.0	uA	1-3
G37	PROM2	29	2.7	0.0	65.0	uA	1-3
G38	PROM1	81	2.7	0.0	65.0	uA	1-3
G39	PROM0	30	2.7	0.0	45.0	uA	1-3
G40	-UDS	28	2.7	0.0	70.0	uA	1-3
G41	-LDS	23					
G42	READ	22	2.7	0.0	70.0	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G43	DTACKI	51	2.7	0.0	100.0	uA	1-3
G44	6 MHZ B	9	2.7	0.0	200.0	uA	1-3
G45	-6 MHZ B	53	2.7	0.0	400.0	uA	1-3
G46	-6 MHZ C	12	2.7	0.0	180.0	uA	1-3
G47	-LD EN	100	2.7	0.0	40.0	uA	1-3
G48	CLK	88	"	"	"	"	"
G49	SELB	44	2.7	0.0	40.0	uA	1-3
G50	-TI1	63	2.7	-2.95	-1.92	mA	1-3
G51	-TI2	70	2.7	-2.95	-2.22	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G51
Power supply current limit	0.913 A	NA	G1-G51
Forcing voltage, (Vih)	2.7 V	± 135 mV	G1-G51

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G51
low, (VL)	0.0 V	± 100 mV	G1-G51
Input current measurement	NA	± 1.0 uA	G1-G21
	NA	± 2.4 uA	G22
	NA	± 1.25 uA	G23-G31
	NA	± 5.0 uA	G32
	NA	± 2.5 uA	G33
	NA	± 4.4 uA	G34
	NA	± 3.0 uA	G35
	NA	± 4.0 uA	G36
	NA	± 3.25 uA	G37,G38
	NA	± 2.25 uA	G39
	NA	± 3.5 uA	G40-G42
	NA	± 5.0 uA	G43
	NA	± 10.0 uA	G44
	NA	± 20.0 uA	G45
	NA	± 9.0 uA	G46
	NA	± 2.0 uA	G47-G49
	NA	± 51.5 uA	G50
	NA	± 36.5 uA	G51

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	A0	45	0.4	-440.0	0.0	uA	1-3
H2	A1	48					
H3	A2	96					
H4	A3	91					
H5	B0	46					
H6	B1	49					
H7	B2	98					
H8	B3	43					
H9	-OE	82	0.4	-440.0	0.0	uA	1-3
H10	-DTACK MUX 0	38	0.4	-400.0	0.0	uA	1-3
H11	-DTACK MUX 1	37					
H12	-DTACK MUX 2	36					
H13	-DTACK MUX 3	35					
H14	-DTACK MUX 4	80					
H15	-DTACK MUX 7	34					
H16	-DTACK MUXEN	41	0.4	-400.0	0.0	uA	1-3
H17	D0	86	0.4	-400.0	-30.0	uA	1-3
H18	D1	42					
H19	D2	93					
H20	D3	95					
H21	-RST	87	0.4	-400.0	-30.0	uA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H22	-AS	26	0.4	-937.0	-516.0	uA	1-3
H23	DELAY SEL 0	77	0.4	-250.0	0.0	uA	1-3
H24	DELAY SEL 1	76					
H25	DELAY SEL 2	79					
H26	DELAY SELEN	90					
H27	CNT0	85					
H28	CNT1	84					
H29	CNT2	83					
H30	CNT3	32					
H31	PROM3	78	0.4	-250.0	0.0	uA	1-3
H32	-DTACK MUX 5	21	0.4	-760.0	-60.0	uA	1-3
H33	PROM0	30	0.4	-650.0	0.0	uA	1-3
H34	-LD EN	100	0.4	-800.0	-30.0	uA	1-3
H35	CLK	88	0.4	-630.0	0.0	uA	1-3
H36	SELB	44	0.4	-880.0	0.0	uA	1-3
H37	-DTACK MUX 6	33	0.4	-1.34	-0.55	mA	1-3
H38	CNT EN	97	0.4	-1.26	-0.06	mA	1-3
H39	PROM2	29	0.4	-1.05	-0.03	mA	1-3
H40	PROM1	81	0.4	-1.05	-0.03	mA	1-3
H41	-UDS	28	0.4	-2.44	-1.06	mA	1-3
H42	-LDS	23	0.4	-2.44	-1.06	mA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H43	READ	22	0.4	-2.44	-1.06	mA	1-3
H44	-6 MHZ C	12	0.4	-4.76	-0.5	mA	1-3
H45	-TI2	70	0.4	-13.7	-5.72	mA	1-3
H46	-TI1	63	0.5	-51.12	-6.78	mA	1-3
H47	REM	24	0.5	-4.0	-0.5	mA	1-3
H48	-BEN	62	0.5	-2.0	-1.0	mA	1-3
H49	DTACKI	51	0.5	-3.2	-0.5	mA	1-3
H50	6 MHZ B	9	0.5	-8.0	-0.5	mA	1-3
H51	-6 MHZ B	53	0.5	-16.0	-0.5	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H51
Power supply current limit	0.913 A	NA	H1-H51
Forcing voltage, (Vil)	0.4 V 0.5 V	± 20 mV ± 25 mV	H1-H45 H46-H51

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	H1-H51
low, (VL)	0.0 V	± 100 mV	H1-H51
Input current measurement	NA	± 22.0 uA	H1-H9
	NA	± 20.0 uA	H10-H16
	NA	± 18.5 uA	H17-H21
	NA	± 21.0 uA	H22
	NA	± 12.5 uA	H23-H31
	NA	± 35.0 uA	H32
	NA	± 32.5 uA	H33
	NA	± 38.5 uA	H34
	NA	± 31.5 uA	H35
	NA	± 44.0 uA	H36
	NA	± 39.5 uA	H37
	NA	± 60.0 uA	H38
	NA	± 51.0 uA	H39,H40
	NA	± 69.0 uA	H41-H43
	NA	± 213.0 uA	H44
	NA	± 0.4 mA	H45
	NA	± 2.22 mA	H46
	NA	± 175.0 uA	H47
	NA	± 50.0 uA	H48
	NA	± 135.0 uA	H49
	NA	± 0.38 mA	H50
	NA	± 0.78 mA	H51

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	Q0	75	-0.4	2.5	4.5	V	1-3
J2	Q1	39					
J3	Q2	89					
J4	Q3	92					
J5	TC	27	-0.4	2.5	4.5	V	1-3
J6	-CUDS	3	-1.0	2.5	4.5	V	1-3
J7	-CLDS	5					
J8	READL1	66					
J9	READL2	61					
J10	-REMREQ	73					
J11	DTACK	54	-1.0	2.5	4.5	V	1-3
J12	Y0	47	-1.0	2.4	4.5	V	1-3
J13	Y1	50					
J14	Y2	99					
J15	Y3	94	-1.0	2.4	4.5	V	1-3
J16	DS EN	71	-1.95	2.4	4.5	V	1-3
J17	DTACK EN	18	-1.95	2.4	4.5	V	1-3
J18	-XDS	74	-0.375	2.5	4.5	V	1-3
J19	-ADDL	2	-0.95	2.5	4.5	V	1-3
J20	-DTACK	7					
J21	-DSW	4	-0.95	2.5	4.5	V	1-3

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J22	-BES	58	-0.95	2.5	4.5	V	1-3
J23	-BENB	16	-0.85	2.5	4.5	V	1-3
J24	BENB	13	-0.9	2.5	4.5	V	1-3
J25	SOCS	19	-0.69	2.5	4.5	V	1-3
J26	ADDL	52	-0.875	2.5	4.5	V	1-3
J27	BES	60	-0.925	2.5	4.5	V	1-3
J28	PU	72	-2.0	2.4	2.6	V	1-3

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J28
Power supply current limit	0.913 A	NA	J1-J28

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Ioh)	-0.4 mA	± 4.0 uA	J1-J5
	-1.0 mA	± 10.0 uA	J6-J15
	-1.95 mA	± 19.5 uA	J16,J17
	-0.375 mA	± 3.75 uA	J18
	-0.95 mA	± 9.5 uA	J19-J22
	-0.85 mA	± 8.5 uA	J23
	-0.9 mA	± 9.0 uA	J24
	-0.69 mA	± 6.9 uA	J25
	-0.895 mA	± 8.75 uA	J26
	-0.925 mA	± 9.25 uA	J27
	-2.0 mA	± 2.0 uA	J28
Pattern voltage high, (VH)	3.0 V	± 100 mV	J1-J28
low, (VL)	0.0 V	± 100 mV	J1-J28
Output voltage measurement	NA	± 100 mV	J1-J11
	NA	± 105 mV	J12-J17
	NA	± 100 mV	J18-J27
	NA	± 10 mV	J28

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

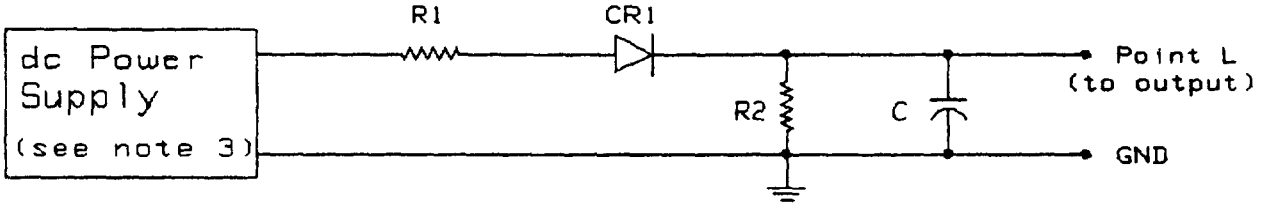
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NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(d) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(e) C not required for any other tests.

3. The load power supply voltage shall be 2.1 V ± 1 percent for the output disable time test and 2.5 V ± 1 percent for all other tests.

FIGURE 6. Logic load circuit.

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NOTES: - Continued.

4. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are listed below.

5. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,58	2632	18.4	0.5
3,5,54,61,66,73	2500	20.0	0.5
4,7	2632	18.0	0.5
13	2778	16.0	0.5
16	2941	14.0	0.5
27,39,75,89,92	6250	4.0	0.4
47,50,94,99	2400	12.0	0.4
52	2857	15.75	0.5
74	6667	3.75	0.4
18,71	1231	6.0	0.4
60	2703	17.75	0.5
72	1200	NA	NA
19	3623	6.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K1	Q0	75	4.0	0.0	0.4	V	1-3
K2	Q1	39					
K3	Q2	89					
K4	Q3	92					
K5	TC	27	4.0	0.0	0.4	V	1-3
K6	Y0	47	12.0	0.0	0.4	V	1-3
K7	Y1	50					
K8	Y2	99					
K9	Y3	94	12.0	0.0	0.4	V	1-3
K10	-XDS	74	3.75	0.0	0.4	V	1-3
K11	DS EN	71	6.0	0.0	0.5	V	1-3
K12	DTACK EN	18	6.0	0.0	0.5	V	1-3
K13	-ADDL	2	18.4	0.0	0.5	V	1-3
K14	-BES	58	18.4	0.0	0.5	V	1-3
K15	-DTACK	7	18.0	0.0	0.5	V	1-3
K16	-DSW	4	18.0	0.0	0.5	V	1-3
K17	ADDL	52	15.8	0.0	0.5	V	1-3
K18	BES	60	17.75	0.0	0.5	V	1-3
K19	-REMREQ	73	20.0	0.0	0.5	V	1-3
K20	DTACK	54	20.0	0.0	0.5	V	1-3
K21	-CUDS	3	20.0	0.2	0.5	V	1-3

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K22	-CLDS	5	20.0	0.2	0.5	V	1-3
K23	READL1	66					
K24	READL2	61	20.0	0.2	0.5	V	1-3
K25	BENB	13	16.0	0.2	0.5	V	1-3
K26	-BENB	16	14.0	0.2	0.5	V	1-3
K27	SOCS	19	6.6	0.2	0.5	V	1-3

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K27
Power supply current limit	0.913 A	NA	K1-K27
Forcing current, (Iol)	4.0 mA	± 40.0 uA	K1-K5
	12.0 mA	± 120.0 uA	K6-K9
	3.75 mA	± 37.5 uA	K10
	6.0 mA	± 60.0 uA	K11,K12
	18.4 mA	± 184.0 uA	K13,K14
	18.0 mA	± 180.0 uA	K15,K16
	15.8 mA	± 158.0 uA	K17
	17.75 mA	± 178.0 uA	K18
	20.0 mA	± 200.0 uA	K19-K24
	16.0 mA	± 160.0 uA	K25
	14.0 mA	± 140.0 uA	K26
	6.6 mA	± 66.0 uA	K27

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K27 K1-K27
Output voltage measurement	 NA NA NA	 ± 20 mV ± 25 mV ± 15 mV	 K1-K10 K11-K20 K21-K27

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	Y0	47	2.7	-20	20	uA	1-3
L2	Y1	50					
L3	Y2	99					
L4	Y3	94	2.7	-20	20	uA	1-3

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L4
Power supply current limit	0.913 A	NA	L1-L4
Forcing voltage, (Voh)	2.7 V	± 135 mV	L1-L4
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	L1-L4 L1-L4
Output current measurement	NA	± 2 uA	L1-L4

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M1	Y0	47	0.4	-20	20	uA	1-3
M2	Y1	50					
M3	Y2	99					
M4	Y3	94	0.4	-20	20	uA	1-3

NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M4
Power supply current limit	0.913 A	NA	M1-M4
Forcing voltage, (Vol)	0.4 V	± 20 mV	M1-M4
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	M1-M4 M1-M4
Output current measurement	NA	± 2 uA	M1-M4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:59Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31	30.4	609.0	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	0.913 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 28.9 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-3
P2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	0.913 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 150 pF (see figure 6). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 7.

3. Abbreviations used are defined in 6.1.

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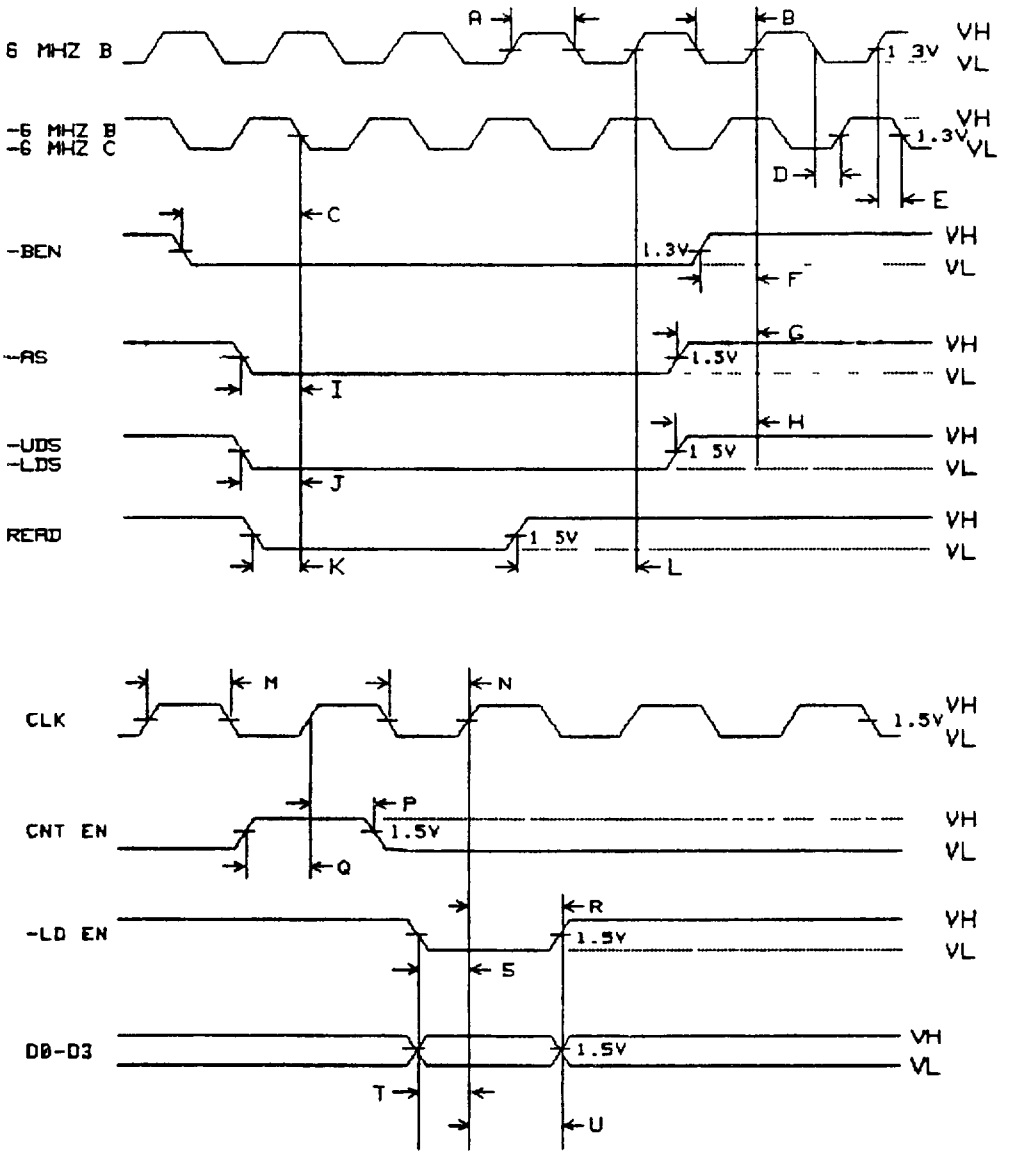


FIGURE 7a. Input timing parameters for table XV.

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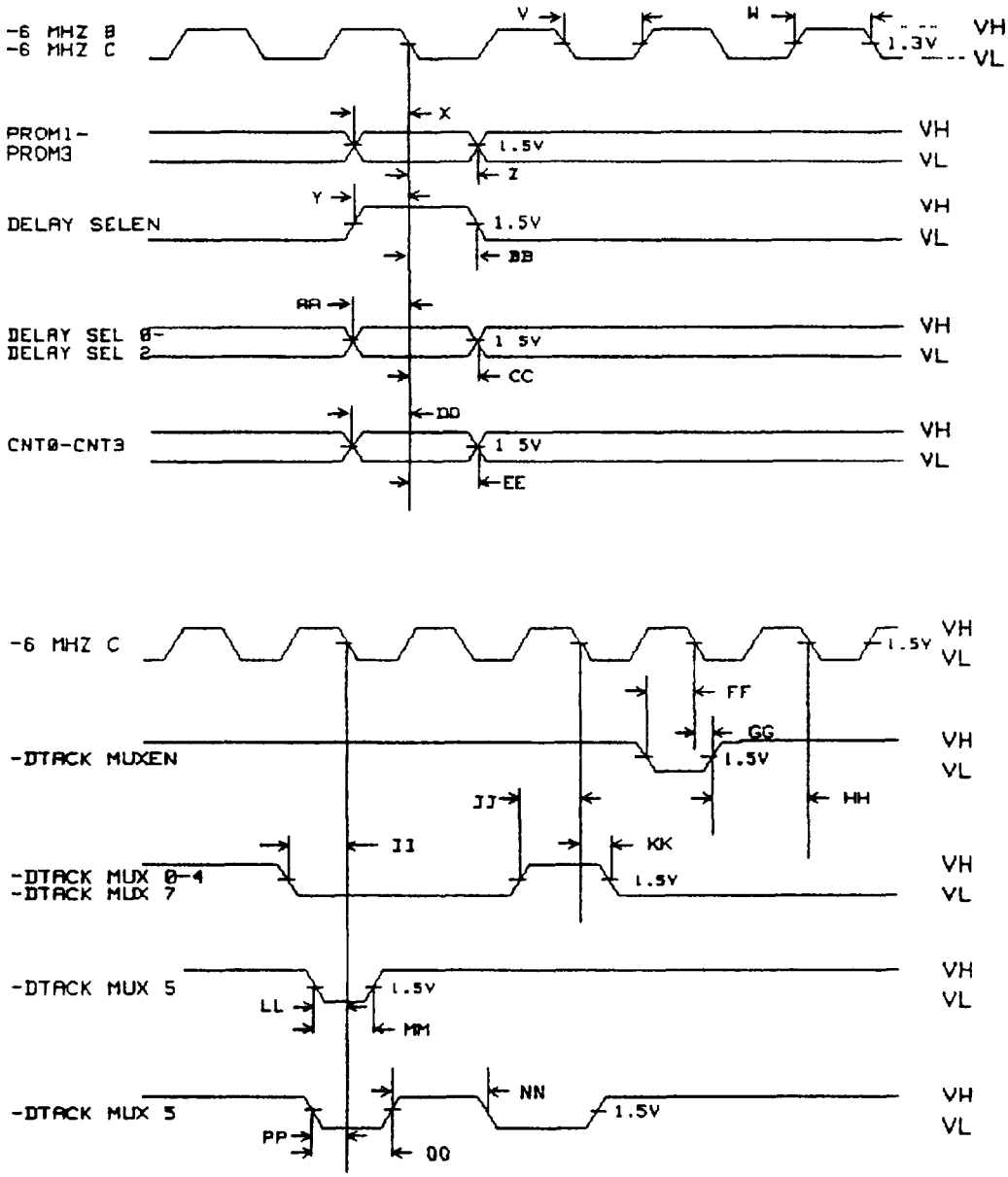


FIGURE 7b. Input timing parameters for table XV.

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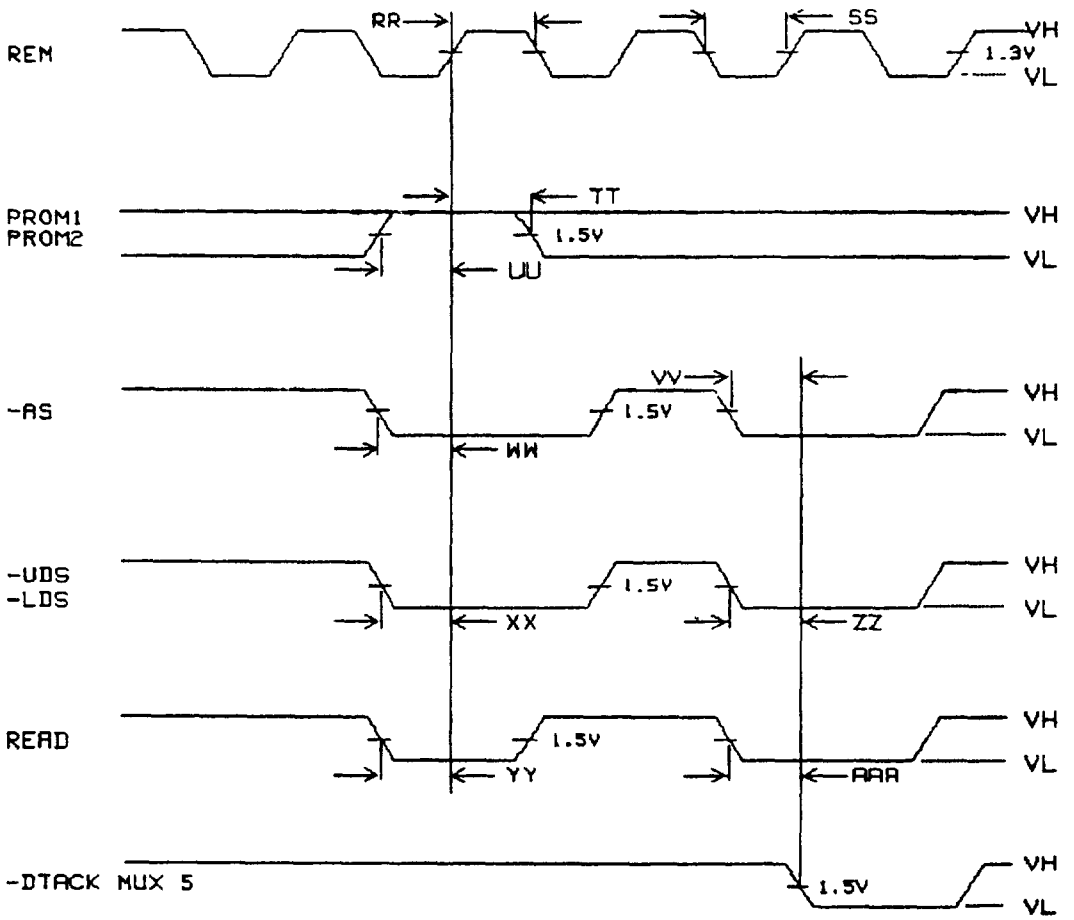


FIGURE 7c. Input timing parameters for table XV.

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NOTES:

1. Input parameters:

A = 77 nS ± 7.7 nS.		AA = 64 nS min.	
B = 77 nS ± 7.7 nS.		BB = 0 nS min.	
C = 30 nS min.		CC = 0 nS min.	
D = 4 nS max.		DD = 64 nS min.	
E = 4 nS max.		EE = 96 nS min.	
F = 21 nS min.		FF = 96 nS min.	
G = 153 nS min.		GG = 0 nS min.	
H = 153 nS min.		HH = 81 nS min.	
I = 153 nS min.		II = 72 nS min.	
J = 153 nS min.		JJ = 81 nS min.	
K = 153 nS min.		KK = 0 nS min.	
L = 153 nS min.		LL = 112 nS min.	
M = 62 nS ± 6.2 nS.		MM = 250 nS min.	
N = 62 nS ± 6.2 nS.		NN = 250 nS min.	
P = 10 nS min.		PP = 114 nS min.	
Q = 25 nS min.		QQ = 250 nS min.	
R = 10 nS min.		RR = 250 nS min.	
S = 25 nS min.		SS = 250 nS min.	
T = 20 nS min.		TT = 0 nS min.	
U = 10 nS min.		UU = 23 nS min.	
V = 77 nS ± 7.7 nS.		VV = 75 nS min.	
W = 77 nS ± 7.7 nS.		WW = 0 nS min.	
X = 112 nS min.		XX = 0 nS min.	
Y = 64 nS min.		YY = 0 nS min.	
Z = 97 nS min.		ZZ = 75 nS min.	
		AAA = 75 nS min.	

2. Pattern voltages shall be as specified in the appropriate test tolerance table.

3. Transition times (as measured between the 0.7 V and 2.0 V points) shall not exceed 20 nS.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-7
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-7
Q3	Dynamic test, qualification maximum pattern rate minimum power supply	1-7
Q4	Dynamic test, qualification maximum pattern rate maximum power supply	1-7

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	0.913 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 105 mV	Q1,Q3
high, (Voh)	NA	± 155 mV	Q2,Q4
low, (Vol)	NA	± 20 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: 2.4 V $\leq$ Voh $\leq$ Vcc
Low level: 0.0 V $\leq$ Vol $\leq$ 0.4 V

3. The load circuit for this test is specified on figure 6.

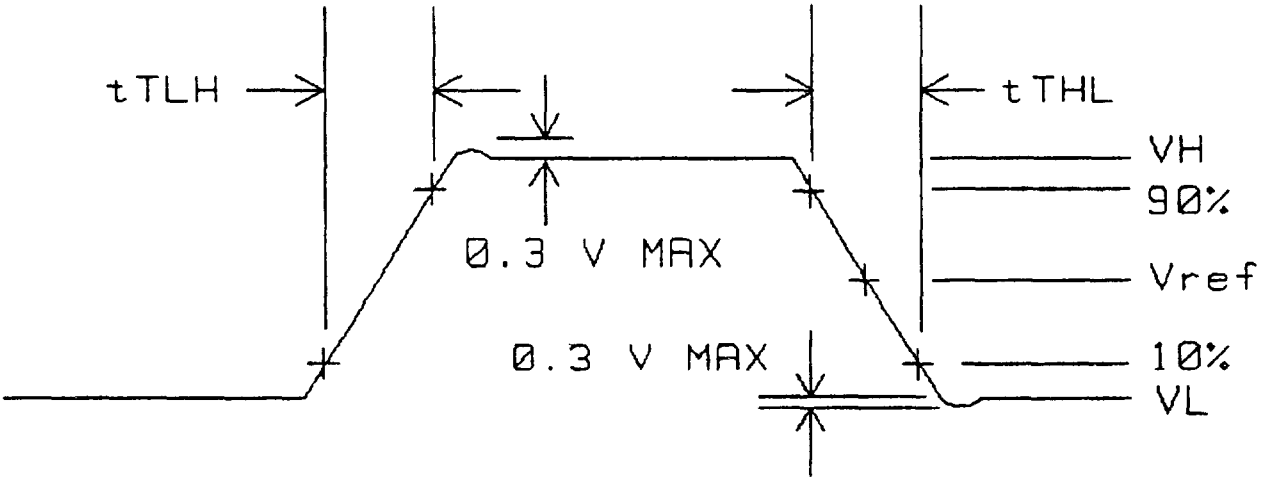
4. All of the input waveforms shall meet the requirements shown on figure 8.

5. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

6. Input/Output (I/O) waveforms are as referenced on figure 9.

7. Abbreviations used are defined in 6.1.

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NOTES:

1. VH and VL are as specified in the test tolerance table.
2. Vref = 1.5 V.
3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
9,12,24,51,53,62	2.5 ± 1.5 nS	2.5 ± 1.5 nS
21,22,23,26,28,29,30,32,33, 34,35,36,37,38,41,42,43,44, 45,46,48,49,76,77,78,79,80, 81,82,83,84,85,86,87,88,90, 91,93,95,96,97,98,100	15.0 ± 1.5 nS	6.0 ± 1.5 nS

FIGURE 8. Sample input waveform for table XVI.

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TABLE XVII. Dynamic parameters.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
122	1	6 MHZ B PWH	77.0	-	77.0	-	nS	9n
123	2	6 MHZ B PWL	77.0	-	77.0	-	nS	9n
	3	6 MHZ B FMAX	6.5	-	6.5	-	MHz	
125	4	-6 MHZ B PWH	77.0	-	77.0	-	nS	9n
124	5	-6 MHZ B PWL	77.0	-	77.0	-	nS	9n
	6	-6 MHZ B FAMX	6.5	-	6.5	-	MHz	
127	7	-6 MHZ C PWH	77.0	-	77.0	-	nS	9n
126	8	-6 MHZ C PWL	77.0	-	77.0	-	nS	9n
	9	-6 MHZ C FMAX	6.5	-	6.5	-	MHz	
128 1/	10	-6 MHZ B (HL) 6 MHZ B (LH) skew	-4.0	4.0	-4.0	4.0	nS	9n
129 1/	11	-6 MHZ B (LH) to 6 MHZ B (HL) skew	-4.0	4.0	-4.0	4.0	nS	9n
131 1/	12	-6 MHZ B (LH) to -6 MHZ C (LH) skew	-4.0	4.0	-4.0	4.0	nS	9n
130 1/	13	-6 MHZ B (HL) to -6 MHZ C (HL) skew	-4.0	4.0	-4.0	4.0	nS	9n
117	14	REM PWH	250.0	-	250.0	-	nS	9m
121	15	REM PWL	250.0	-	250.0	-	nS	9m
	16	REM FMAX	2.0	-	2.0	-	MHz	
93	17	CLK PWH	62.0	-	62.0	-	nS	9k

1/ Measure with respect to -6 MHZ B.

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
94	18	CLK PWL	62.0	-	62.0	-	nS	9k
	19	CLK FMAX	8.0	-	8.0	-	MHz	
83	20	-DTACK MUX 5 PWH	250.0	-	250.0	-	nS	9i
85	21	-DTACK MUX 5 PWL	250.0	-	250.0	-	nS	9i
	22	-DTACK MUX 5 FMAX	2.0	-	2.0	-	MHz	
39	23	-BEN to -BENB, tPLH	-	28.0	-	25.0	nS	9d
40	24	-BEN to -BENB, tPHL	-	28.0	-	25.0	nS	9d
38	25	-BEN to BENB, tPLH	-	15.0	-	13.0	nS	9d
37	26	-BEN to BENB, tPHL	-	13.0	-	11.5	nS	9d
19 32	27	-BEN to SOCS, tPLH	-	39.0	-	34.0	nS	9b, 9c
6	28	-BEN (LH) to 6 MHZ B, setup time	28.0	-	25.0	-	nS	9a
15 26	29	-BEN (HL) -6 MHZ B, setup time	39.0	-	34.0	-	nS	9b, 9c
17 30	30	-AS to SOCS, tPLH	-	56.0	-	44.5	nS	9b, 9c
18 31	31	-AS to SOCS, tPHL	-	60.0	-	46.5	nS	9b, 9c

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
1	32	-AS (HL) to 6 MHZ B, setup time (-BEN=1)	35.0	-	27.0	-	nS	9a
4	33	-AS (LH) to 6 MHZ B, setup time (-BEN=1)	41.0	-	31.0	-	nS	9a
13 24 27	34	-AS to -6 MHZ B, setup time (-BEN=0)	69.0	-	55.5	-	nS	9b, 9c
2	35	-AS to 6 MHZ B, hold time (-BEN=1)	0.0	-	0.0	-	nS	9a
14 25	36	-AS to -6 MHZ B, hold time (-BEN=0)	0.0	-	0.0	-	nS	9b, 9c
50 62	37	-AS to -DTACK, tPLH	-	58.0	-	43.0	nS	9f, 9g
48 60	38	-AS to DTACK tPHL	-	56.0	-	41.0	nS	9f, 9g
51	39	-AS to -DSW, tPLH	-	75.0	-	59.0	nS	9f
52 65	40	-AS to -CUDS, tPLH	-	82.0	-	64.5	nS	9f, 9g
52 65	41	-AS to -CLDS, tPLH	-	82.0	-	64.5	nS	9f, 9g
134	42	-AS to -REMREQ, tPLH	-	54.0	-	42.0	nS	9m

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
81	43	-AS to -6 MHZ C, setup time	173.0	-	130.0	-	nS	9i
21	44	-AS to BES (-BES=0), tPHL	-	75.0	-	59.5	nS	9b
23	45	-AS to BES (-BEN=0), tPLH	-	90.0	-	72.5	nS	9b
17 30	46,47	-UDS, -LDS to SOCS, tPLH	-	56.0	-	44.5	nS	9b, 9c
18 31	48,49	-UDS, -LDS to SOCS, tPHL	-	60.0	-	47.0	nS	9b, 9c
3	50,51	-UDS, -LDS (HL) to 6 MHZ B, setup time (-BEN=1)	35.0	-	27.0	-	nS	9a
5	52,53	-UDS, -LDS (LH) to 6 MHZ B, setup time (-BEN=1)	41.0	-	31.0	-	nS	9a
16 28 29	54,55	-UDS, -LDS to -6 MHZ B, setup time (-BEN=0)	69.0	-	55.5	-	nS	9b, 9c
2	56,57	-UDS, -LDS to 6 MHZ B, hold time (-BEN=1)	0.0	-	0.0	-	nS	9a
14	58,59	-UDS, -LDS to -6 MHZ B, hold time (-BEN=0)	0.0	-	0.0	-	nS	9b
50 62	60,61	-UDS, -LDS to -DTACK, tPLH	-	58.0	-	43.0	nS	9f, 9g

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
48 60	62,63	-UDS, -LDS to DTACK, tPHL	-	56.0	-	41.0	nS	9f, 9g
65 52	64	-UDS to -CUDS, tPLH	-	24.0	-	20.5	nS	9f, 9g
65 52	65	-LDS to -CLDS, tPLH	-	24.0	-	20.5	nS	9f, 9g
134	66, 67	-UDS, -LDS to -REMREQ, tPLH	-	54.0	-	42.0	nS	9m
81	68, 69	-UDS, -LDS to -6 MHZ C, setup time	173.0	-	130.0	-	nS	9i
21	70, 71	-UDS, -LDS to BES (-BEN=0), tPHL	-	75.0	-	59.5	nS	9b
23	72, 73	-UDS, -LDS to -BES (-BEN=0), tPLH	-	90.0	-	72.5	nS	9b
17 30	74	READ to SOCS, tPLH	-	56.0	-	44.5	nS	9b, 9c
18 31	75	READ to SOCS, tPHL	-	60.0	-	46.5	nS	9b, 9c
3	76	READ (HL) to 6 MHZ B, setup time (-BEN=1)	35.0	-	27.0	-	nS	9a
5	77	READ (LH) to 6 MHZ B, setup time (-BEN=1)	41.0	-	31.0	-	nS	9a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
16 28	78	READ to -6 MHZ B, setup time (-BEN=0)	69.0	-	55.5	-	nS	9b, 9c
2	79	READ to 6 MHZ B, hold time (-BEN=1)	0.0	-	0.0	-	nS	9a
14	80	READ to -6 MHZ B, hold time (-BEN=0)	0.0	-	0.0	-	nS	9b
50 62	81	READ to -DTACK, tPLH	-	58.0	-	43.0	nS	9f, 9g
48 60	82	READ to DTACK, tPHL	-	56.0	-	41.0	nS	9f, 9g
51	83	READ to -DSW, tPLH	-	44.0	-	34.5	nS	9f
134	84	READ to -REMREQ, tPLH	-	54.0	-	42.0	nS	9m
81	85	READ to -6 MHZ C, setup time	173.0	-	130.0	-	nS	9i
21	86	READ to BES, tPHL (-BEN=0)	-	75.0	-	59.5	nS	9b
23	87	READ to -BES, tPLH (-BEN=0)	-	90.0	-	72.5	nS	9b
9 20 33	88	-6 MHZ B to BES, tPLH	-	17.0	-	15.0	nS	9a, 9b, 9c
11 22 35	89	-6 MHZ B to -BES, tPHL	-	30.0	-	26.0	nS	9a, 9b, 9c

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
10	90	6 MHZ B to BES, tPHL (-BEN=1)	-	44.0	-	37.5	nS	9a
12	91	6 MHZ B to -BES, tPLH (-BEN = 1)	-	59.0	-	50.5	nS	9a
64	92	-6 MHZ B to ADDL, tPLH	-	17.0	-	15.0	nS	9g
66	93	-6 MHZ B to ADDL, tPHL	-	15.0	-	13.0	nS	9g
68	94	-6 MHZ B to -ADDL, tPLH	-	17.0	-	15.0	nS	9g
67	95	-6 MHZ B to -ADDL, tPHL	-	15.0	-	13.0	nS	9g
59	96	-6 MHZ B to DTACK, tPLH	-	17.0	-	15.0	nS	9g
61	97	-6 MHZ B to -DTACK, tPHL	-	15.0	-	13.0	nS	9g
63 1/	98	-6 MHZ C to -CUDS, tPHL	-	57.0	-	47.5	nS	9g
63 1/	99	-6 MHZ C to -CLDS, tPHL	-	57.0	-	47.5	nS	9g
46	100	6 MHZ B to -DSW, tPHL	-	15.0	-	13.0	nS	9f
36	101	-6 MHZ B to -BES, tPLH 2/	-	35.0	-	29.0	nS	9c

1/ Measure with respect to -6 MHZ B.

2/ When -AS or -UDS, -LDS and READ are negated more than one clock cycle, but less than two clock cycles after DTACK is generated.

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
34	102	-6 MHZ B to BES, tPHL 1/	-	26.0	-	22.0	nS	9c
57	103-105	PROM1-PROM3 to DS EN, tPLH	-	52.0	-	52.0	nS	9g
42 47	106-108	PROM1-PROM3 to DS EN, tPHL	-	56.0	-	56.0	nS	9e, 9f
57	109	DELAY SELEN to DS EN, tPLH	-	52.0	-	52.0	nS	9g
42 47	110	DELAY SELEN to DS EN, tPHL	-	56.0	-	56.0	nS	9e, 9f
57	111	BES to DS EN, tPLH	-	52.0	-	52.0	nS	9g
42 47	112	BES to DS EN, tPHL	-	56.0	-	56.0	nS	9e, 9f
57	113-115	DELAY SEL 0 - DELAY SEL 2 to DS EN, tPLH	-	52.0	-	52.0	nS	9g
42 47	116-118	DELAY SEL 0 - DELAY SEL 2 to DS EN, tPHL	-	56.0	-	56.0	nS	9e, 9f
57	119	CNT1 to DS EN, tPLH	-	52.0	-	52.0	nS	9g
42 47	120	CNT1 to DS EN, tPHL	-	56.0	-	56.0	nS	9e, 9f

1/ When -AS or -UDS, -LDS and READ are negated more than one clock cycle, but less than two clock cycles after DTACK is generated.

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
74 55 69 82	121-123	PROM0-PROM2 to -6 MHZ C, setup time	97.0	-	73.0	-	nS	9g, 9h, 9i
54	124-126	PROM0-PROM2 to -6 MHZ C, hold time	0.0	-	0.0	-	nS	9g
55	127	DELAY SELEN to -6 MHZ C, setup time	79.0	-	76.0	-	nS	9g
54	128	DELAY SELEN to -6 MHZ C, hold time	0.0	-	0.0	-	nS	9g
55	131-133	DELAY SEL 0 - DELAY SEL 2 to -6 MHZ C, setup time	79.0	-	76.0	-	nS	9g
54	134-136	DELAY SEL 0 - DELAY SEL 2 to -6 MHZ C, hold time	0.0	-	0.0	-	nS	9g
55	137	CNT1 to -6 MHZ C, setup time	79.0	-	76.0	-	nS	9g
54	138	CNT1 to -6 MHZ C, hold time	0.0	-	0.0	-	nS	9g
58	139,140	PROM1, PROM3 to DTACK EN, tPLH (PROM3 H or LH)	-	52.0	-	52.0	nS	9g

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
43 49	141,142	PROM1, PROM3 to DTACK EN, tPHL (PROM3 H or HL)	-	56.0	-	56.0	nS	9g, 9h
56	143,144	PROM1, PROM3 to -6 MHZ B, setup time	79.0	-	76.0	-	nS	9h
54	145,146	PROM1, PROM3 to to -6 MHZ B, hold time	0.0	-	0.0	-	nS	9h
58	147	-XDS to DTACK EN, tPLH	-	52.0	-	52.0	nS	9e
43 49	148	-XDS to DTACK EN, tPLH	-	56.0	-	56.0	nS	9e, 9f
58	149	BES to DTACK EN, tPLH	-	52.0	-	52.0	nS	9g
43 49	150	BES to DTACK EN, tPHL	-	56.0	-	56.0	nS	9e, 9f
58	151-153	DELAY SEL 0 - DELAY SEL 2 to DTACK EN, tPLH	-	52.0	-	52.0	nS	9g
43 49	154-156	DELAY SEL 0 - DELAY SEL 2 to DTACK EN, tPHL	-	56.0	-	56.0	nS	9e, 9f
56	157-159	DELAY SEL 0 - DELAY SEL 2 to -6 MHZ B, setup time	79.0	-	76.0	-	nS	9g

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
54	160-162	DELAY SEL 0 - DELAY SEL 2 to -6 MHZ B, hold time	0.0	-	0.0	-	nS	9g
58	163-166	CNT0-CNT3 to DTACK EN, tPLH	-	52.0	-	52.0	nS	9g
43 49	167-170	CNT0-CNT3 to DTACK EN, tPHL	-	56.0	-	56.0	nS	9e, 9f
56	171-174	CNT0-CNT3 to -6 MHZ B, setup time	79.0	-	76.0	-	nS	9g
54	175-178	CNT0-CNT3 to -6 MHZ B, hold time	0.0	-	0.0	-	nS	9g
132	179	DTACKI to -6 MHZ B, setup time	9.0	-	9.0	-	nS	9o
133	180	DTACKI to -6 MHZ B, hold time	2.0	-	2.0	-	nS	9o
120	181	REM to -REMREQ, tPHL	-	17.0	-	14.0	nS	9m
118	182,183	PROM1, PROM2 to REM, setup time	27.0	-	20.0	-	nS	9m
116	184,185	PROM1, PROM2 to REM, hold time	0.0	-	0.0	-	nS	9m

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
76 79	186-190	-DTACK MUX 0 through -DTACK MUX 4 to -6 MHZ C, setup time	81.0	-	62.0	-	nS	9h
77	191-195	-DTACK MUX 0 through -DTACK MUX 4 to -6 MHZ C, hold time	0.0	-	0.0	-	nS	9h
76 79	196	-DTACK MUX 7 to -6 MHZ C, setup time	81.0	-	62.0	-	nS	9h
77	197	-DTACK MUX 7 to -6 MHZ C, hold time	0.0	-	0.0	-	nS	9h
89	198	-DTACK MUX 6 to -6 MHZ C, setup time	121.0	-	90.0	-	nS	9j
84	199	-DTACK MUX 5 to -6 MHZ C, setup time	123.0	-	92.0	-	nS	9i
75 80 88 92	200	-6 MHZ C to -XDS, tPLH	-	44.0	-	34.0	nS	9h, 9i, 9j
72 78 87 91	201	-6 MHZ C to -XDS, tPHL	-	53.0	-	41.0	nS	9h, 9i, 9j

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
70 71	202	-DTACK MUXEN to -6 MHZ C, setup time	96.0	-	72.0	-	nS	9h
73	203	-DTACK MUXEN to -6 MHZ C, hold time	0.0	-	0.0	-	nS	9h
90	204	-DTACK MUX 6 PWL	25.0	-	25.0	-	nS	9j
53	205,206	READ to READL1, READL2, tPLH	-	24.0	-	20.5	nS	9g
41	207,208	READ to READL1 READL2, tPHL	-	22.0	-	18.5	nS	9e
112	209-212	A0-A3 to Y0-Y3, tPHL	-	39.0	-	27.0	nS	91
113	213-216	A0-A3 to Y0-Y3, tPLH	-	42.0	-	30.0	nS	91
110	217-220	B0-B3 to Y0-Y3, tPHL	-	39.0	-	27.0	nS	91
109	221-224	B0-B3 to Y0-Y3, tPLH	-	42.0	-	30.0	nS	91
107 111	225-228	SELB to Y0-Y3, tPLH	-	46.0	-	33.0	nS	91
108	229-232	SELB to Y0-Y3, tPHL	-	43.0	-	30.0	nS	91
114	233-236	-OE to Y0-Y3, tPZH	-	60.0	-	42.0	nS	91
115	237-240	-OE to Y0-Y3, tPZL	-	57.0	-	39.0	nS	91

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
100	241	CNT EN to CLK, setup time	25.0	-	25.0	-	nS	9k
99	242	CNT EN to CLK, hold time	10.0	-	10.0	-	nS	9k
95	243-246	D0-D3 to CLK, setup time	20.0	-	20.0	-	nS	9k
96	247-250	D0-D3 to CLK, hold time	10.0	-	10.0	-	nS	9k
97	251	-LD EN to CLK, setup time	25.0	-	25.0	-	nS	9k
98	252	-LD EN to CLK, hold time	10.0	-	10.0	-	nS	9k
101	253	-RST PWL time	20.0	-	20.0	-	nS	9k
102	254-257	CLK to Q0-Q3, tPLH	-	53.0	-	41.0	nS	9k
103	258-261	CLK to Q0-Q3, tPHL	-	56.0	-	43.0	nS	9k
136	262	CLK to TC, tPLH	-	68.0	-	52.0	nS	9p
137	263	CLK to TC, tPHL	-	67.0	-	51.0	nS	9p
138	264	CNT EN to TC, tPLH	-	40.0	-	31.0	nS	9p
139	265	CNT EN to TC, tPHL	-	39.0	-	30.0	nS	9p
106	266-269	-RST to Q0-Q3, tPHL	-	57.0	-	44.0	nS	9k

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
44	270	6 MHZ B to -DSW, tPLH	-	17.0	-	15.0	nS	9e
45	271	6 MHZ B to -CLDS, tPLH	-	41.0	-	35.5	nS	9e
45	272	6 MHZ B to -CUDS, tPLH	-	41.0	-	35.5	nS	9e
105	273-276	CLK to Q0-Q3, tPHL (-LD EN =0)	-	54.0	-	41.0	nS	9k
104	277-280	CLK to Q0-Q3, tPHL (-LD EN =0)	-	59.0	-	43.0	nS	9k
7	281	6 MHZ B to SOCS, tPLH (-BEN=1)	-	28.0	-	23.0	nS	9a
8	282	6 MHZ B to SOCS, tPHL (-BEN = 1)	-	29.0	-	24.5	nS	9a
119	284	-AS to REM, setup time	34.0	-	26.0	-	nS	9m
119	285	-UDS to REM, setup time	34.0	-	26.0	-	nS	9m
119	286	-LDS to REM, setup time	34.0	-	26.0	-	nS	9m
119	287	READ to REM, setup time	34.0	-	26.0	-	nS	9m
55	288	PROM3 to -6 MHZ C, setup time	79.0	-	76.0	-	nS	9g
54	289	PROM3 to -6 MHZ C, hold time	0.0	-	0.0	-	nS	9g

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	0 to 60°C		25 ± 5°C Initial limits		Units	Fig
			Min	Max	Min	Max		
86	290	-AS setup time (-DTACK MUX 5)	58.0	-	46.5	-	nS	9i
87	291	READ setup time (-DTACK MUX 5)	58.0	-	46.5	-	nS	9i
86	292	-UDS setup time (-DTACK MUX 5)	58.0	-	46.5	-	nS	9i
86	293	-LDS setup time (-DTACK MUX 5)	58.0	-	46.5	-	nS	9i

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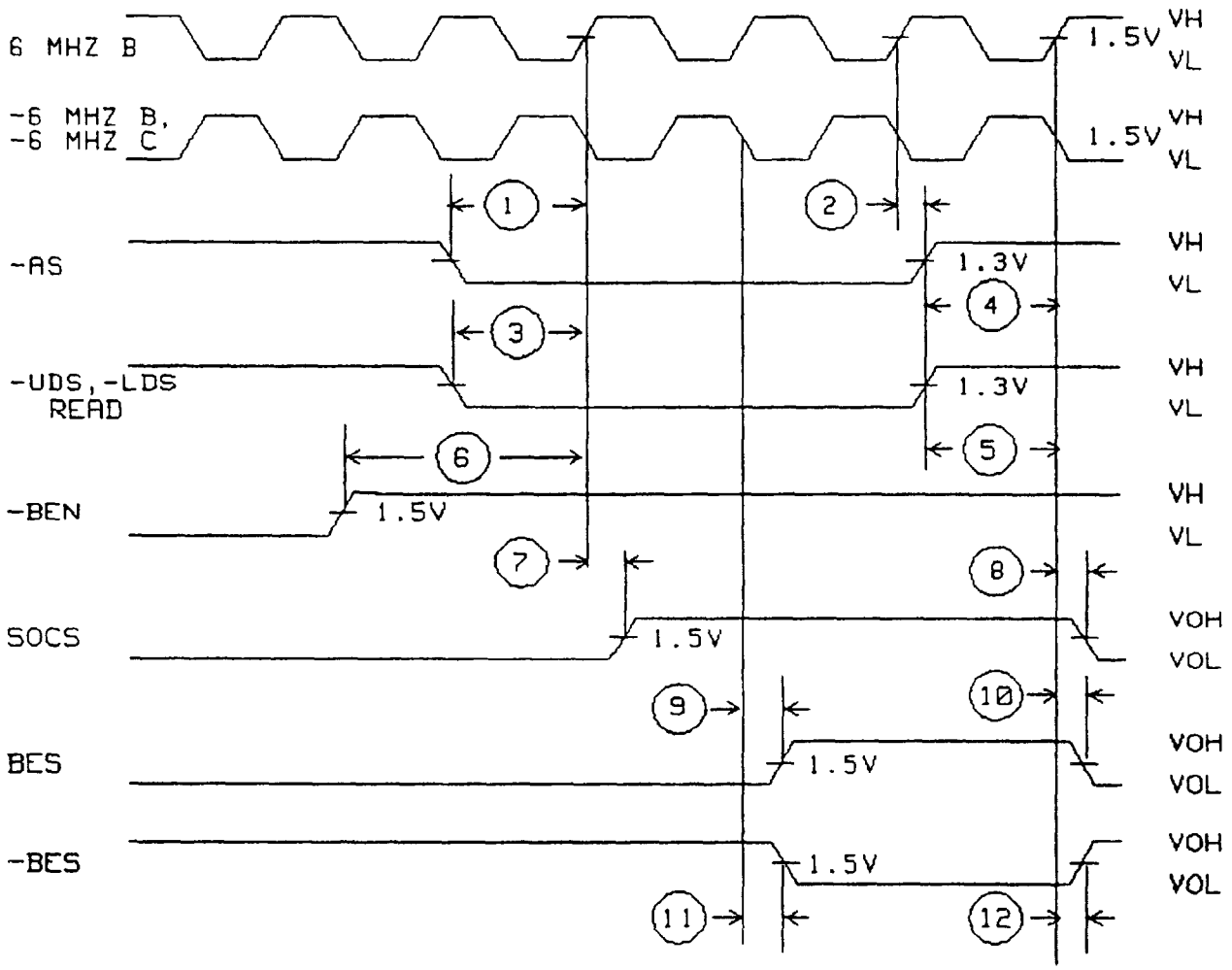


FIGURE 9a. Dynamic waveforms for table XVII.

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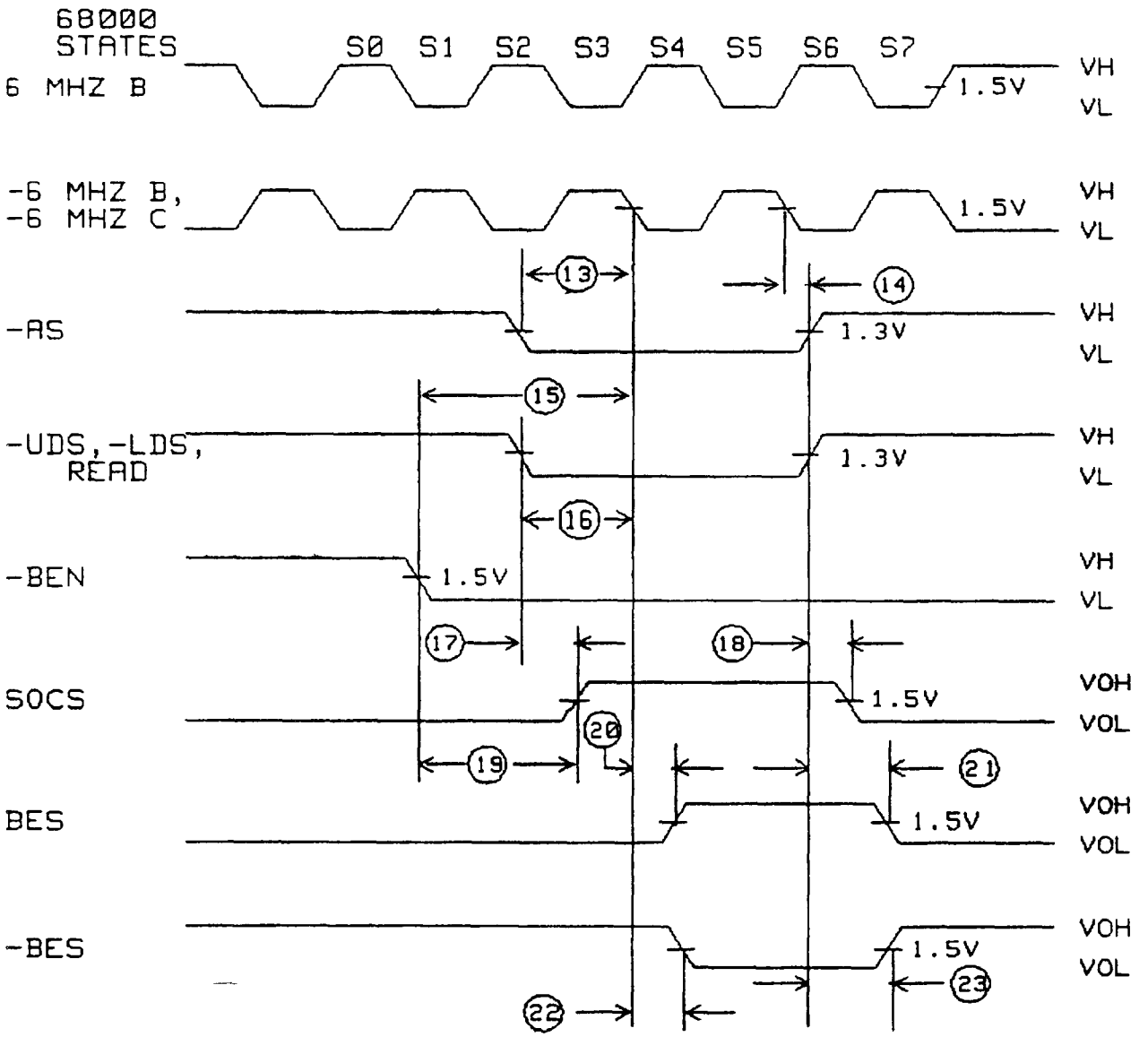


FIGURE 9b. Dynamic waveforms for table XVII.

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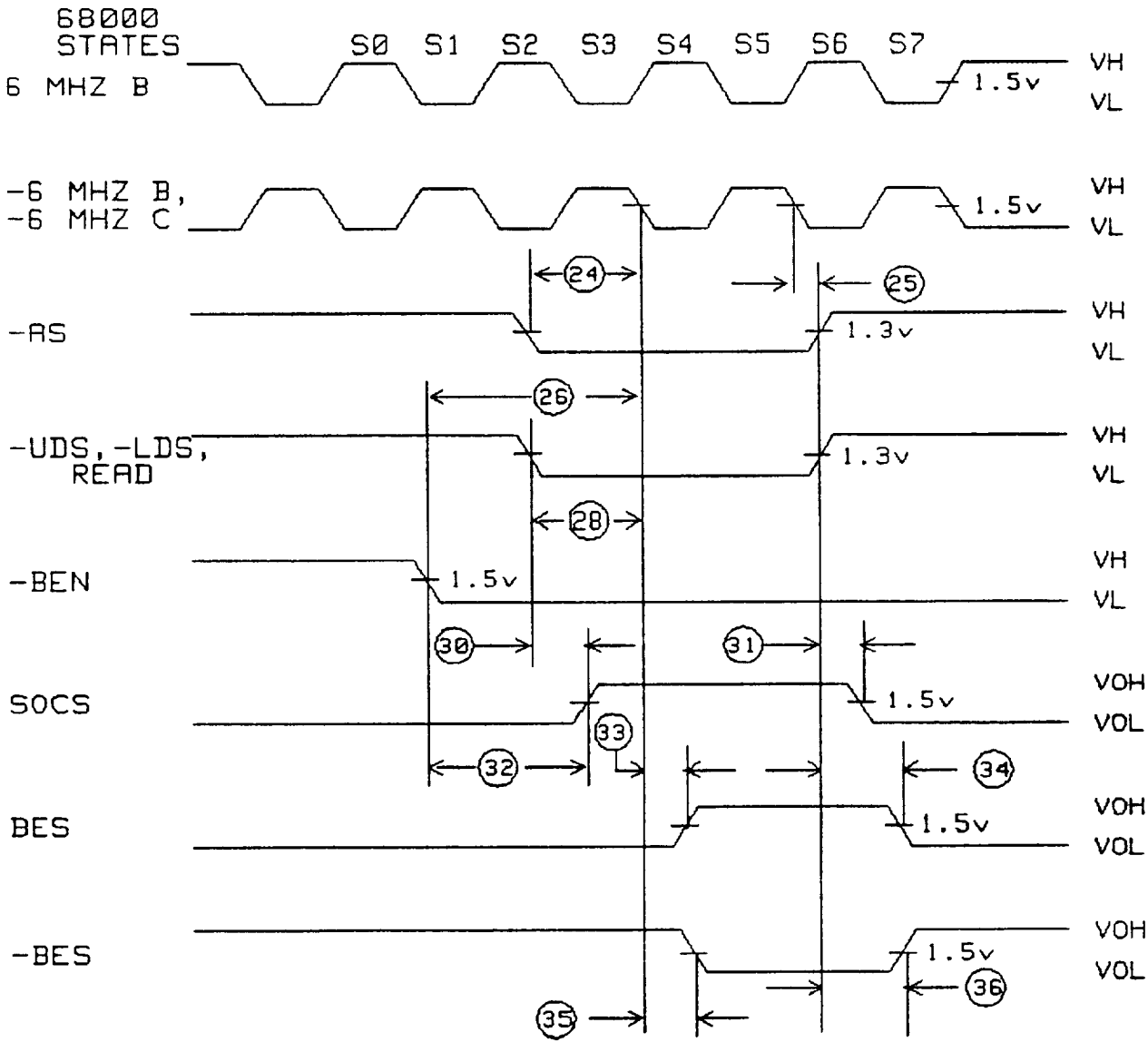


FIGURE 9c. Dynamic waveforms for table XVII.

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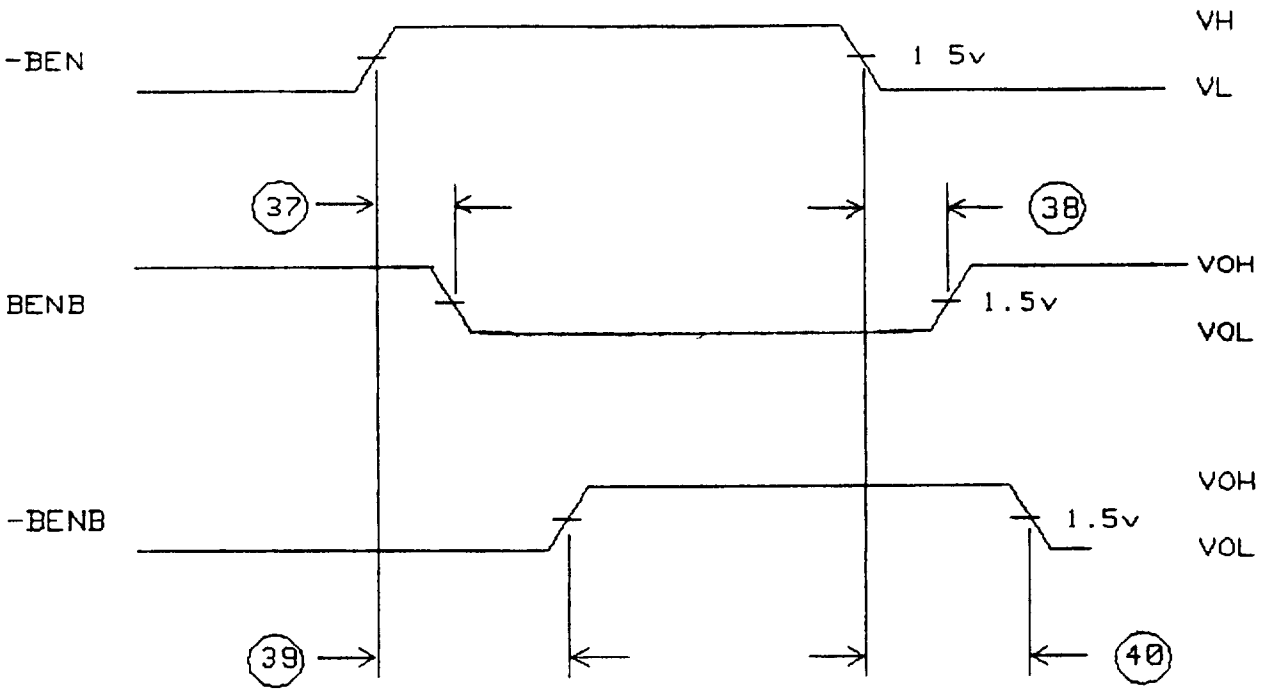


FIGURE 9d. Dynamic waveforms for table XVII.

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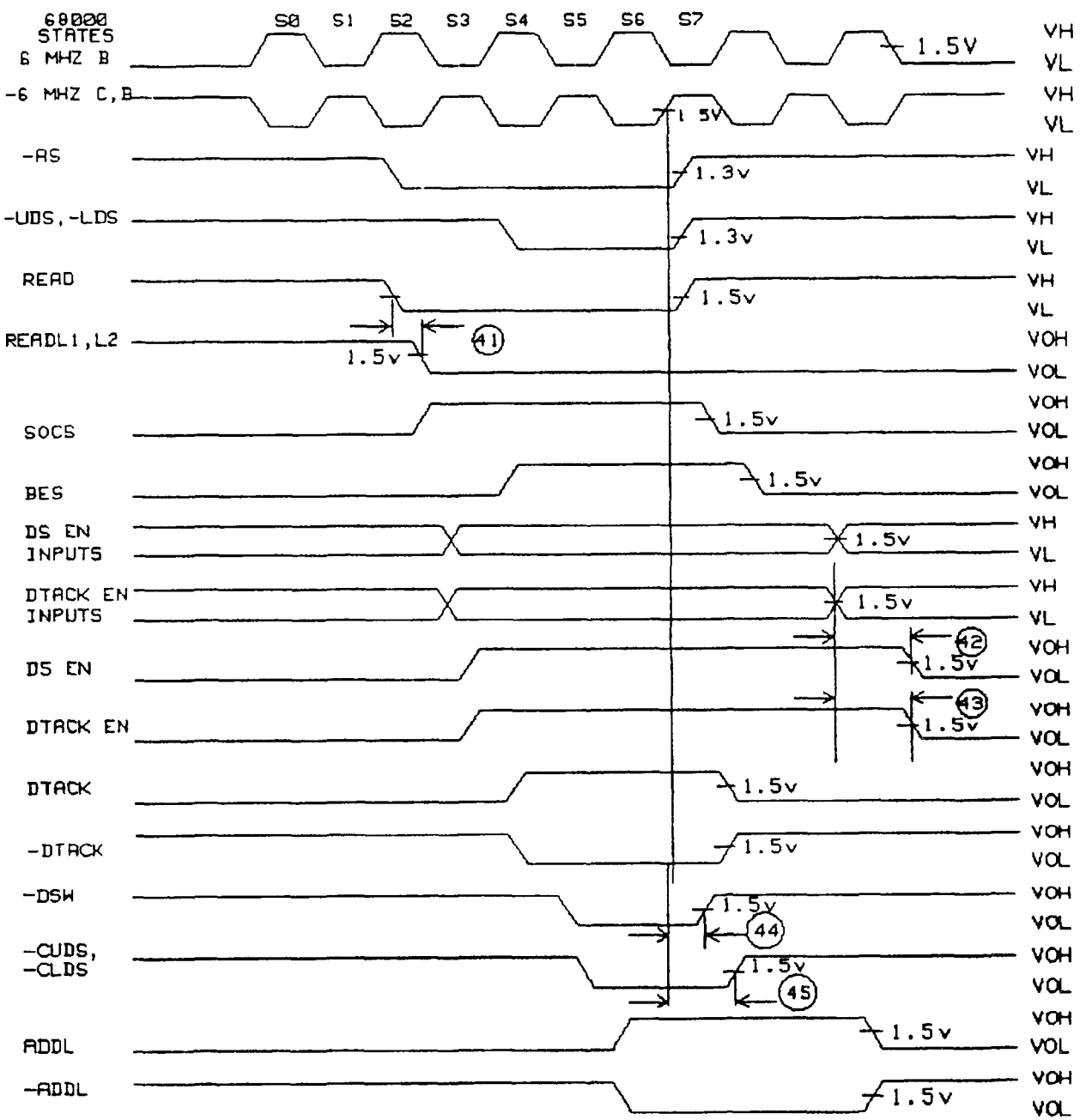


FIGURE 9e. Dynamic waveforms for table XVII.

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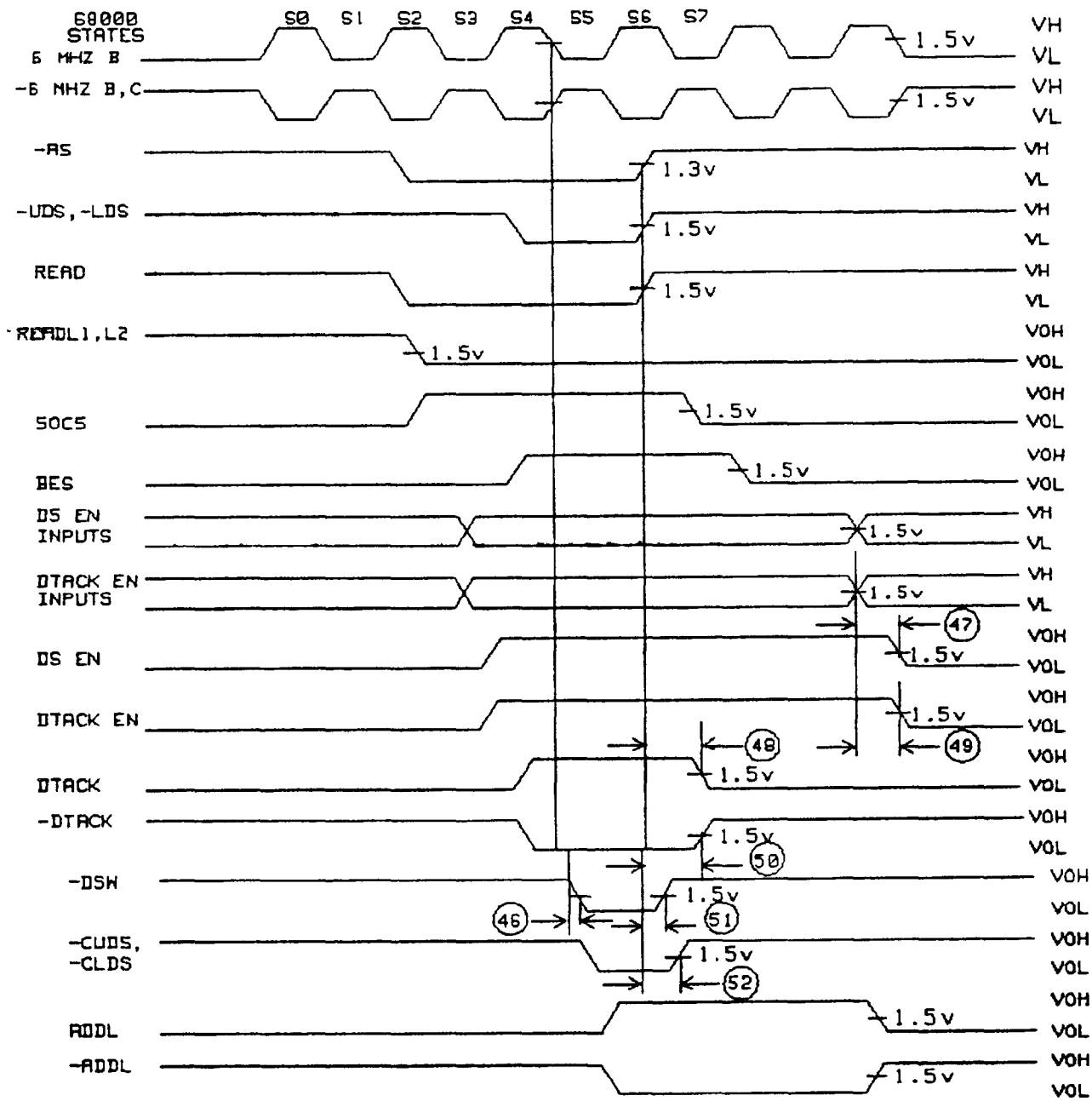


FIGURE 9f. Dynamic waveforms for table XVII.

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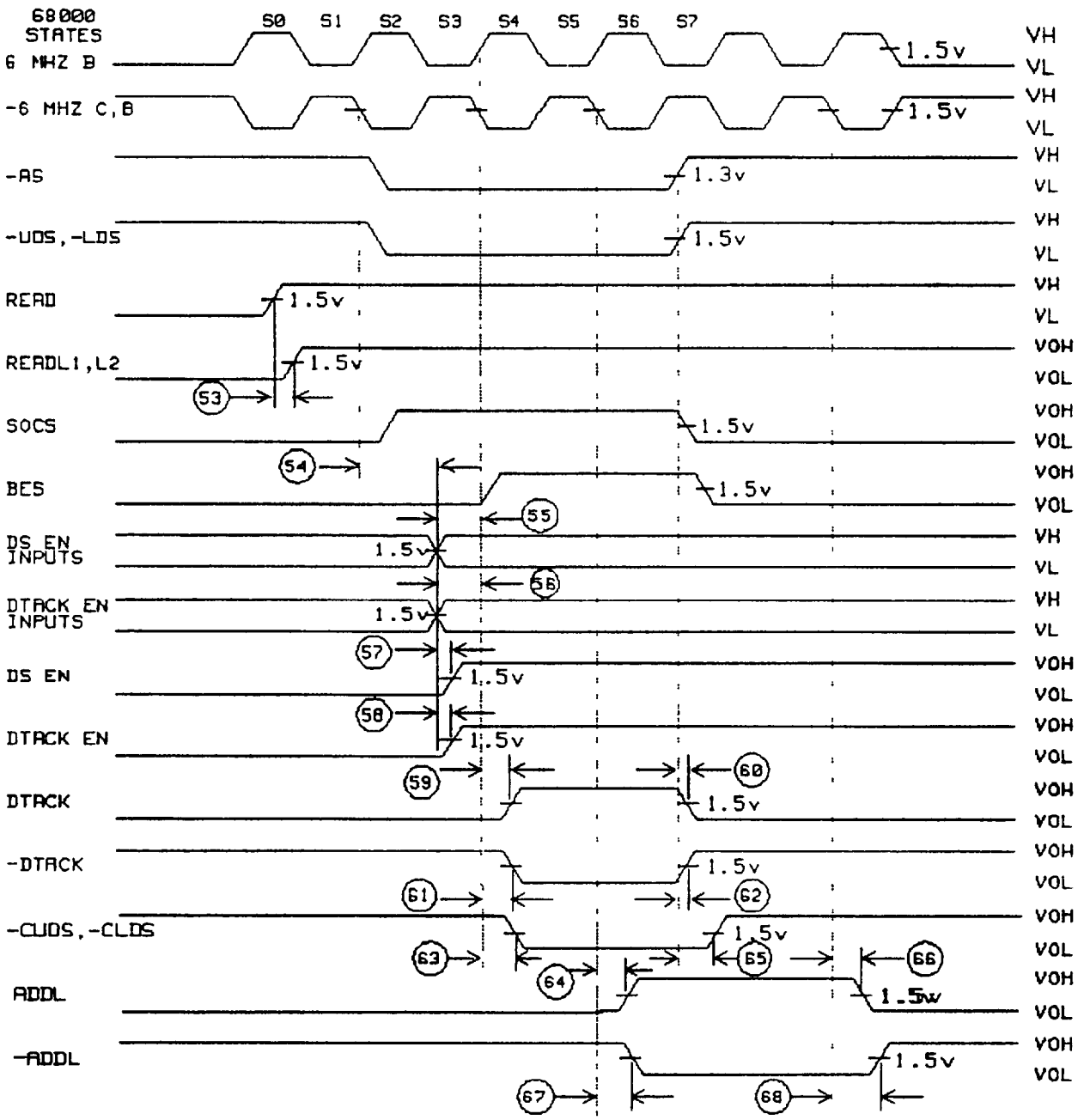


FIGURE 9g. Dynamic waveforms for table XVII.

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Check the source to verify that this is the current version before use.

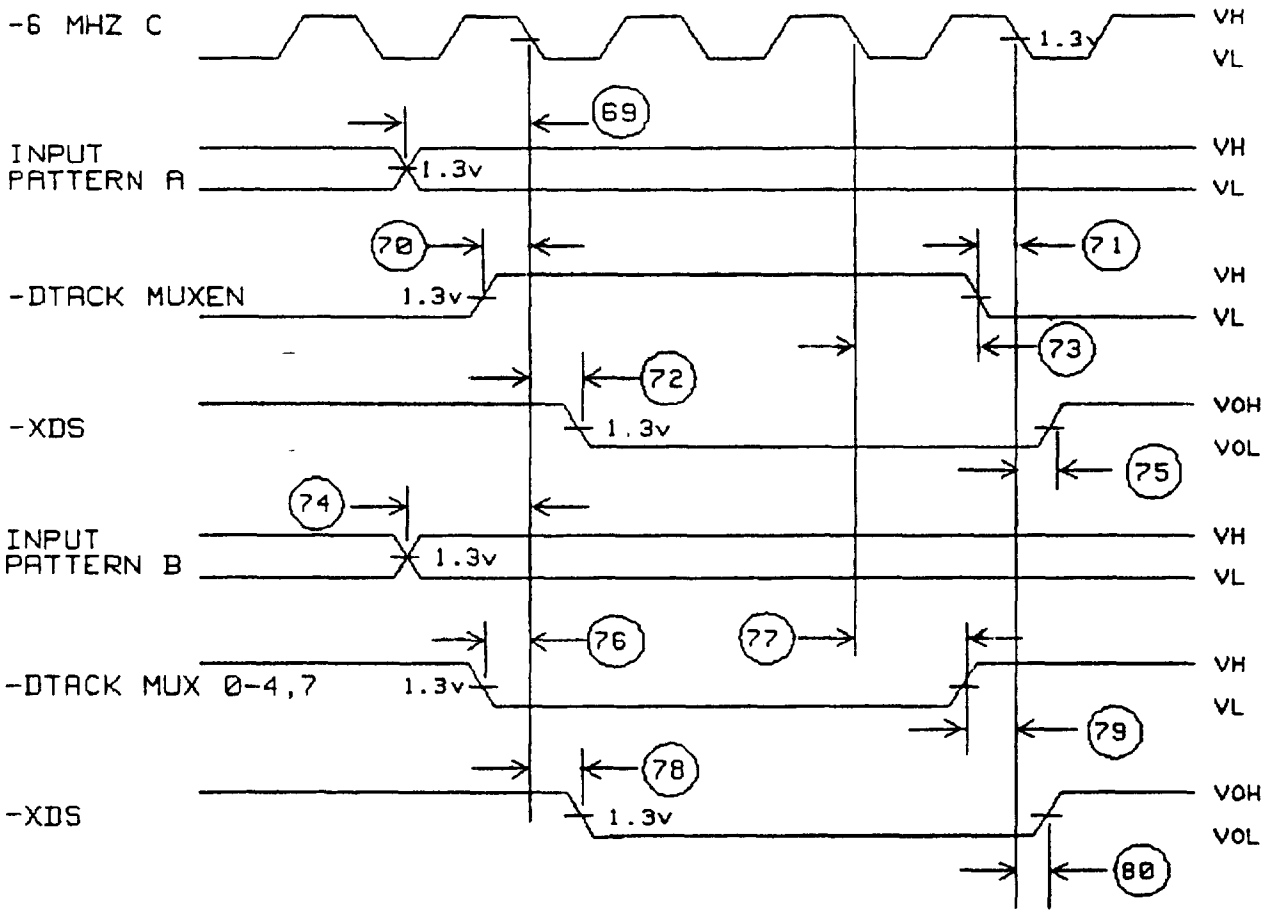


FIGURE 9h. Dynamic waveforms for table XVII.

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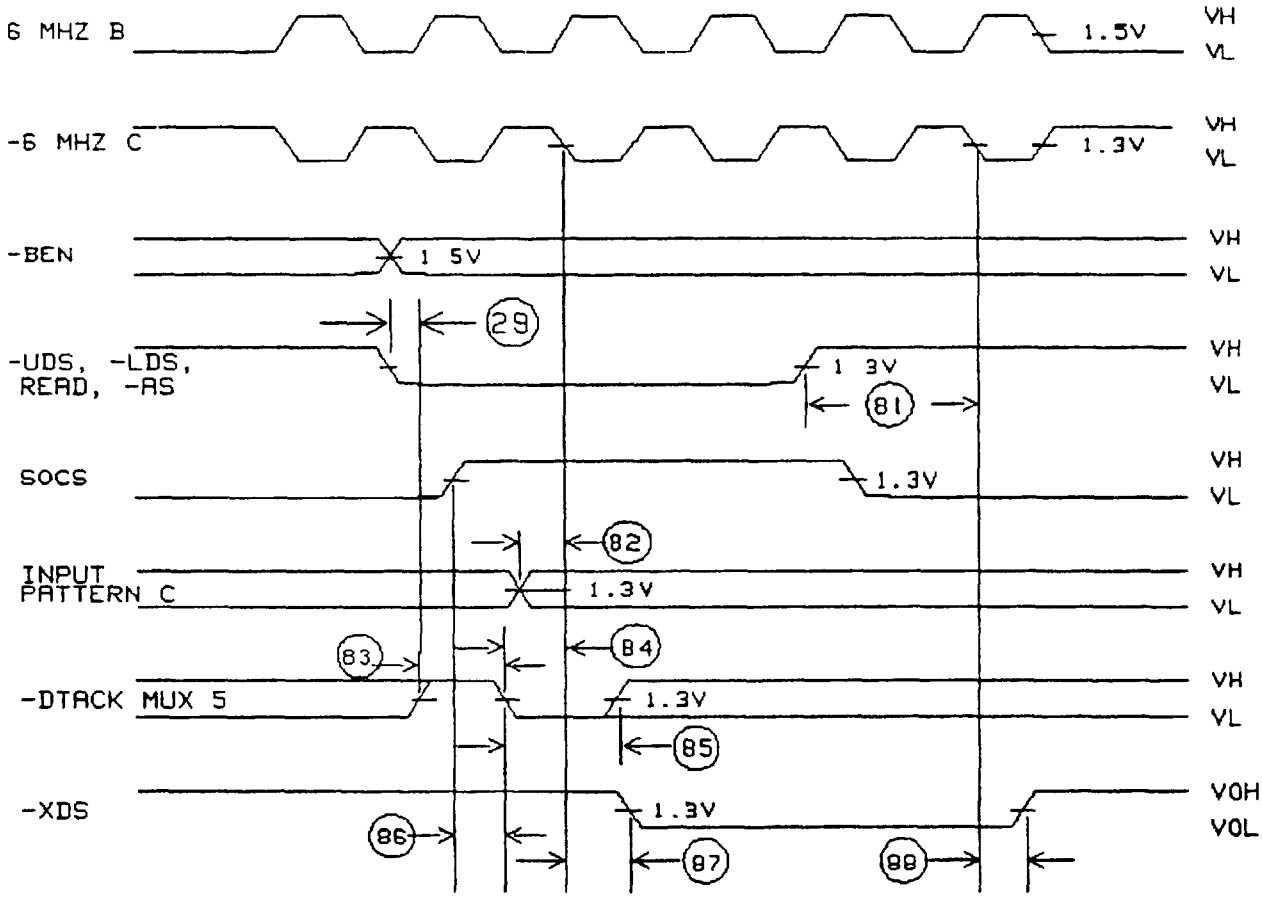


FIGURE 9i. Dynamic waveforms for table XVII.

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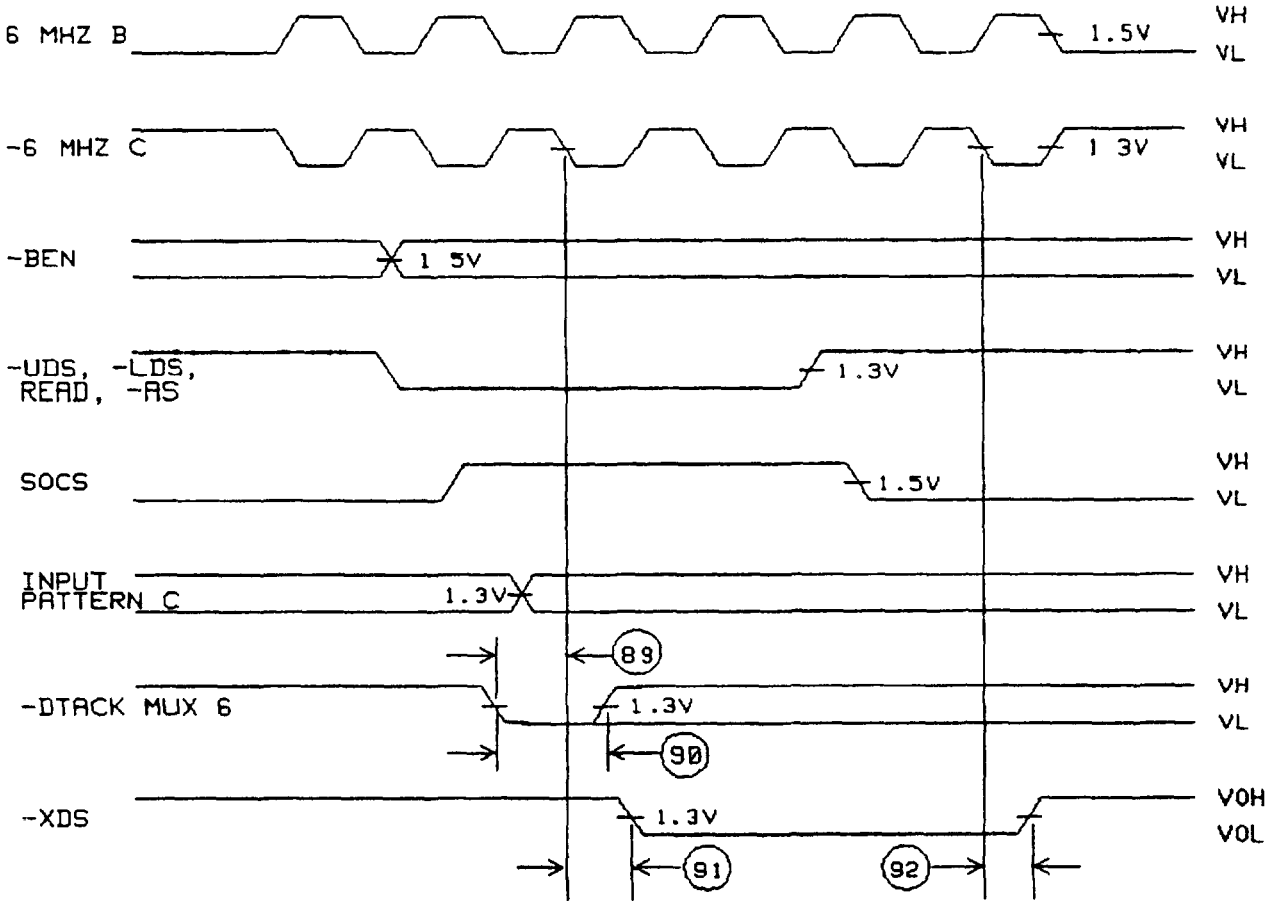


FIGURE 9j. Dynamic waveforms for table XVII.

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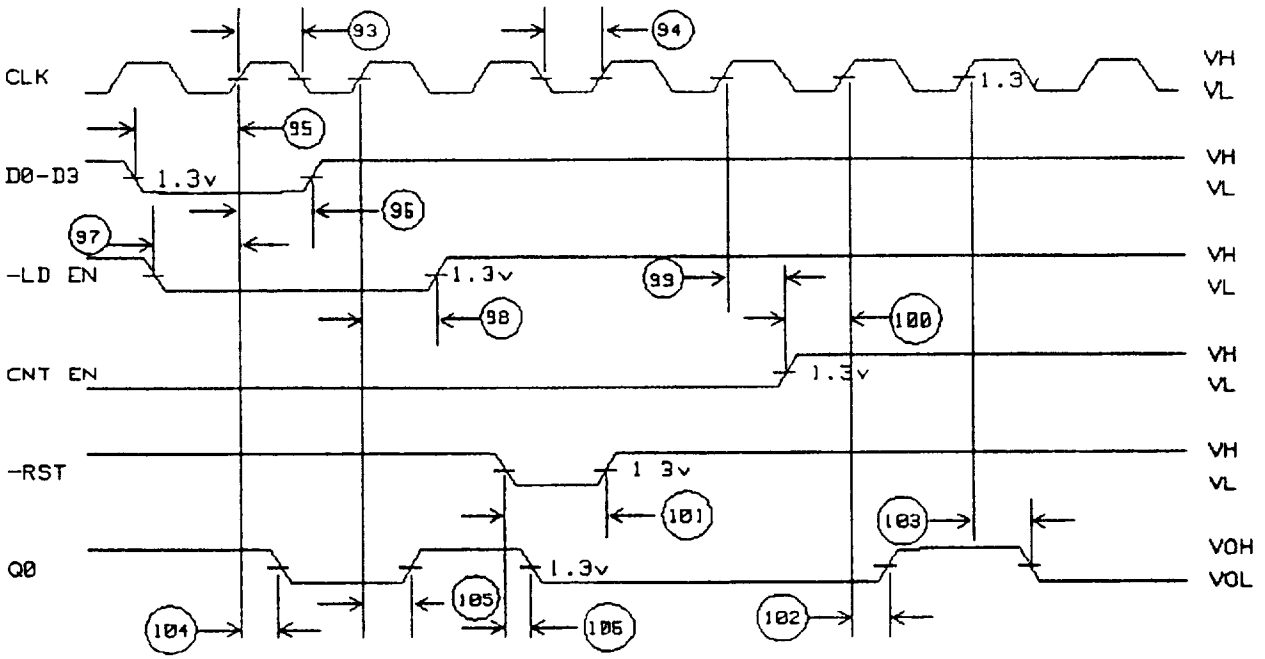


FIGURE 9k. Dynamic waveforms for table XVII.

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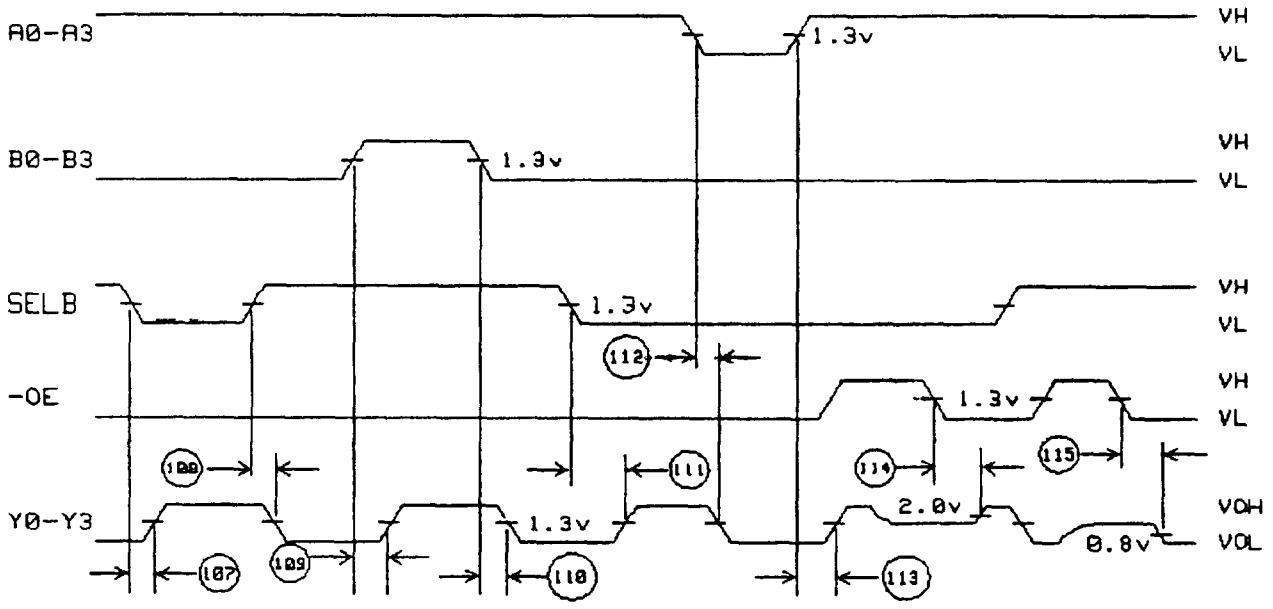


FIGURE 91. Dynamic waveforms for table XVII.

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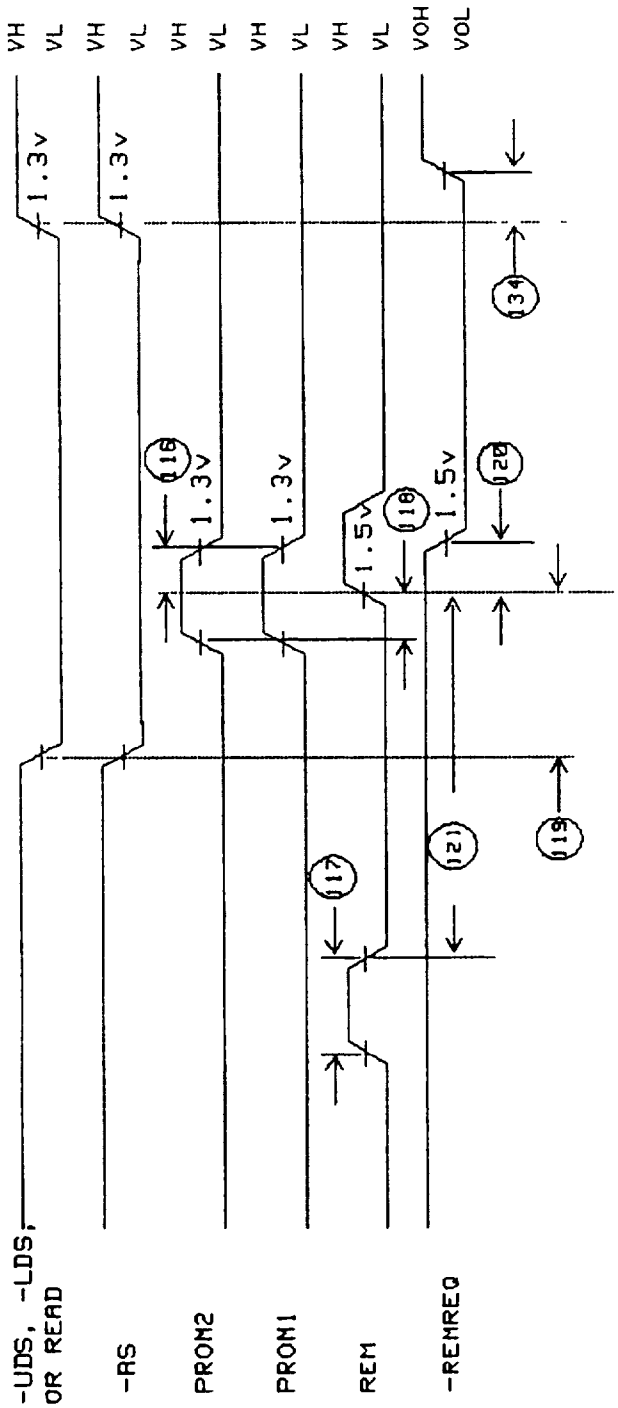


FIGURE 9m. Dynamic waveforms for table XVII.

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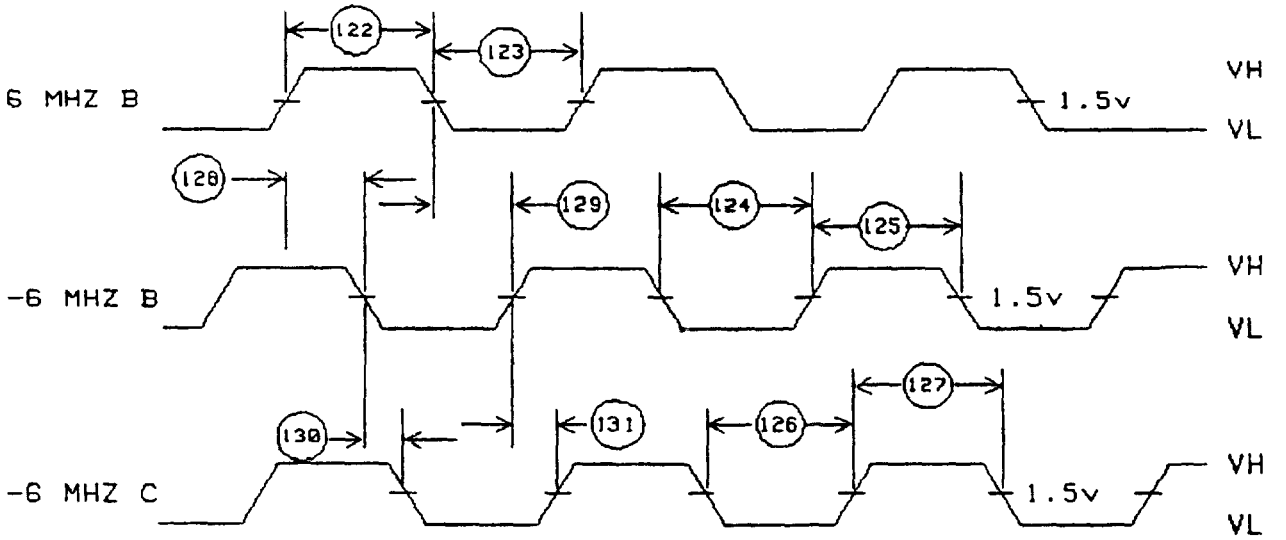


FIGURE 9n. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:59Z
Check the source to verify that this is the current version before use.

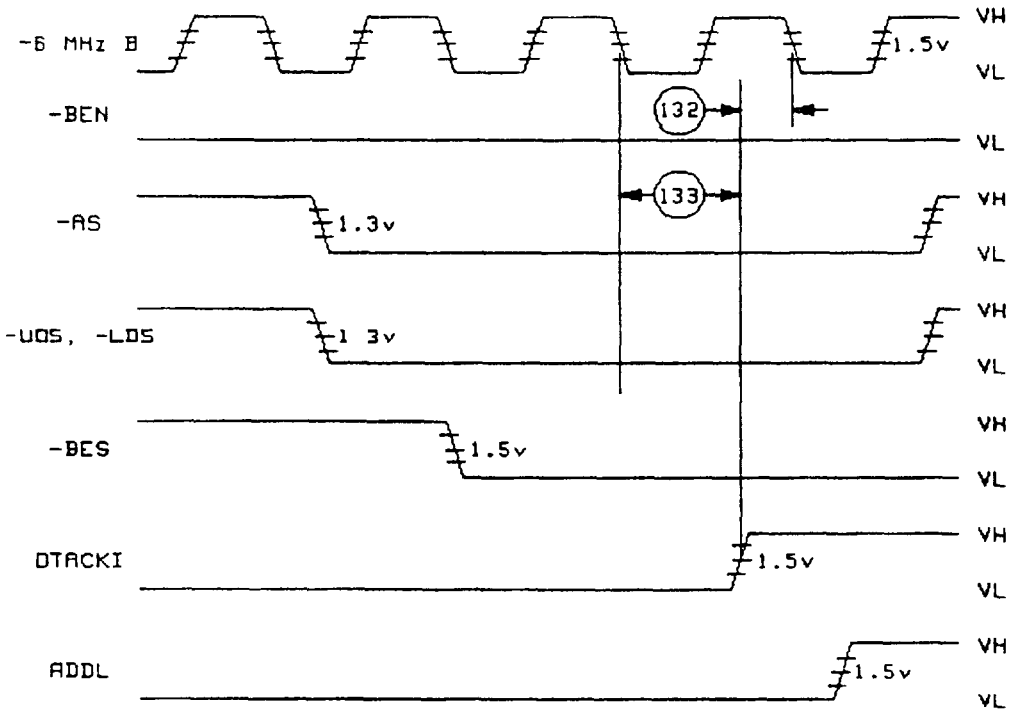


FIGURE 9o. Dynamic waveforms for table XVII.

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Check the source to verify that this is the current version before use.

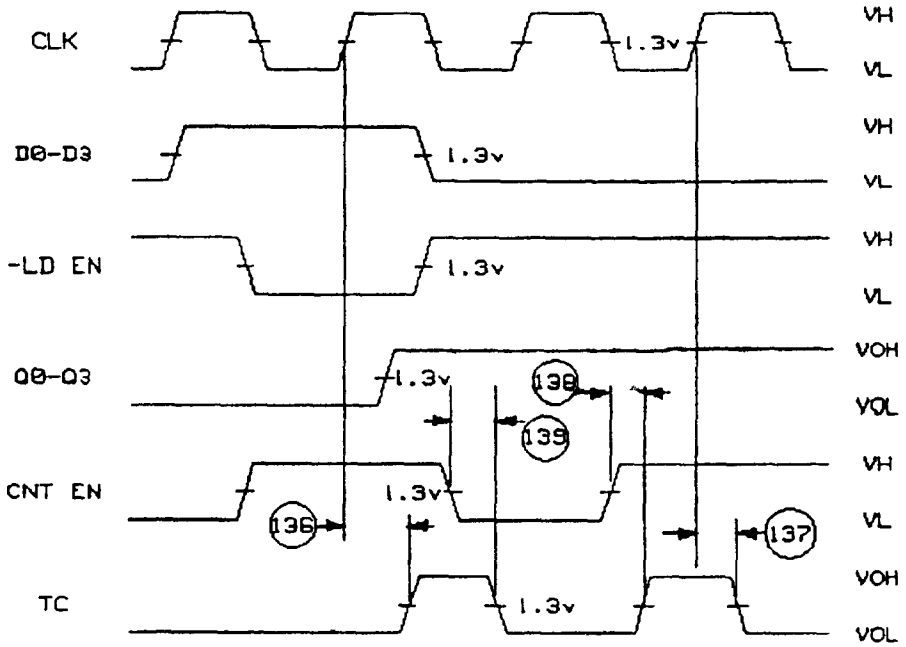


FIGURE 9p. Dynamic waveforms for table XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	4.0	7.6	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 180 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S	PSTA	1	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	0.913 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

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NOTES: - Continued.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 10 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. Module inputs shall not violate the minimum input timing parameters referenced on figure 7.

4. Abbreviations used are defined in 6.1.

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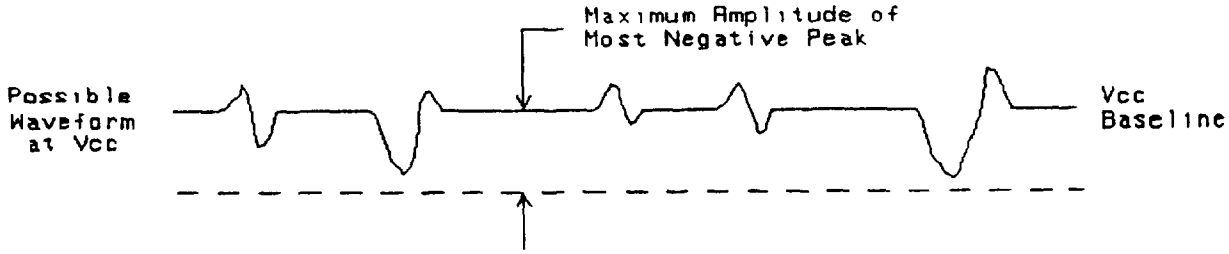


FIGURE 10. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	Q0	75	0	400	mV	1-4	11,12
T2	Q1	39					
T3	Q2	89					
T4	Q3	92					
T5	TC	27					
T6	Y0	47					
T7	Y1	50					
T8	Y2	99					
T9	Y3	94					
T10	-XDS	74					
T11	PU	72	0	400	mV	1-4	11,12
T12	-CLDS	5	0	500	mV	1-4	11,12
T13	READL1	66					
T14	READL2	61					
T15	-REMREQ	73					
T16	DTACK	54					
T17	DS EN	71					
T18	DTACK EN	18					
T19	-ADDL	2					
T20	ADDL	52					
T21	-DSW	4	0	500	mV	1-4	11,12

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T22	BENB	13	0	500	mV	1-4	11,12
T23	-BENB	16					
T24	-DTACK	7					
T25	SOCS	19					
T26	-BES	58					
T27	BES	60					
T28	-CUDS	3	0	500	mV	1-4	11,12

NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	T1-T28
Power supply current limit	0.913 A	NA	T1-T28
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T28 T1-T28
Output crosstalk measurement	 NA NA	 ± 20 mV ± 25 mV	 T1-T11 T12-T28

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. Module inputs shall not violate the minimum input timing parameters referenced on figure 7.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS $\pm$ 200 nS, 50 percent duty cycle.

(d) For each output under test in table XX, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 11.

(e) The maximum probe capacitance is 10 pF.

3. The test circuit shall be as indicated on figure 12. All outputs shall be loaded as specified on figure 6.

4. Abbreviations used are defined in 6.1.

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TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
U1	Q0	75	0	400	mV	1-4	11,12
U2	Q1	39					
U3	Q2	89					
U4	Q3	92					
U5	TC	27					
U6	-XDS	74	0	400	mV	1-4	11,12
U7	-CLDS	5	0	500	mV	1-4	11,12
U8	-CUDS	3					
U9	READL1	66					
U10	READL2	61					
U11	-REMREQ	73					
U12	DTACK	54					
U13	DS EN	71					
U14	DTACK EN	18					
U15	-ADDL	2					
U16	ADDL	52					
U17	-DSW	4					
U18	BENB	13					
U19	-BENB	16					
U20	-DTACK	7					
U21	SOCS	19	0	500	mV	1-4	11,12

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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
U22	-BES	58	0	500	mV	1-4	11,12
U23	BES	60	0	500	mV	1-4	11,12
U24	Y0	47	0	650	mV	1-4	11,12
U25	Y1	50					
U26	Y2	99					
U27	Y3	94	0	650	mV	1-4	11,12

NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	U1-U27
Power supply current limit	0.913 A	NA	U1-U27
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 U1-U27 U1-U27
Output crosstalk measurement	 NA NA NA	 ± 20.0 mV ± 25.0 mV ± 32.5 mV	 U1-U6 U7-U23 U24-U27

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. Module inputs shall not violate the minimum input timing parameters referenced on figure 7.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XXI, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 11.

(e) The maximum probe capacitance is 10 pF.

3. The test circuit shall be as indicated on figure 12. All outputs shall be loaded as specified on figure 6.

4. Abbreviations used are defined in 6.1.

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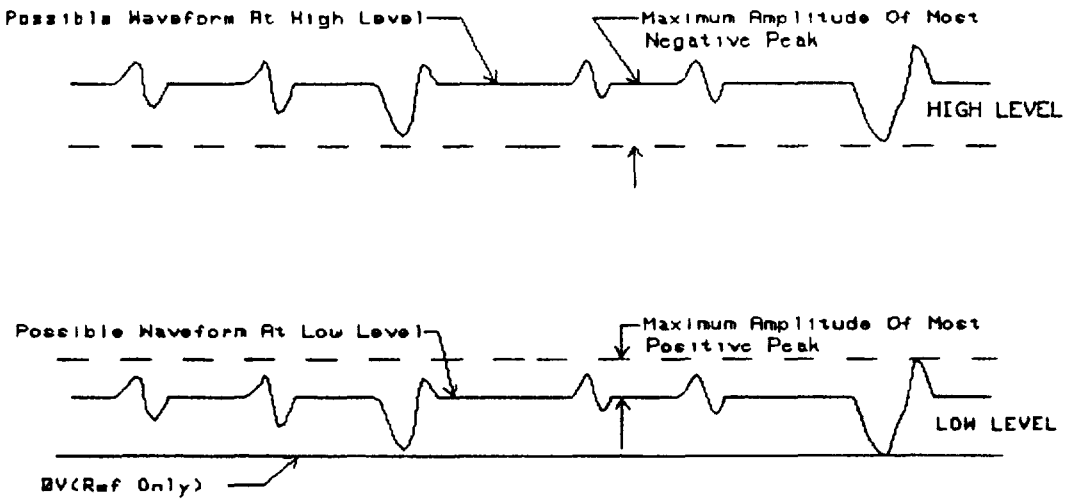


FIGURE 11. Output crosstalk, high and low level, test waveforms for tables XX and XXI.

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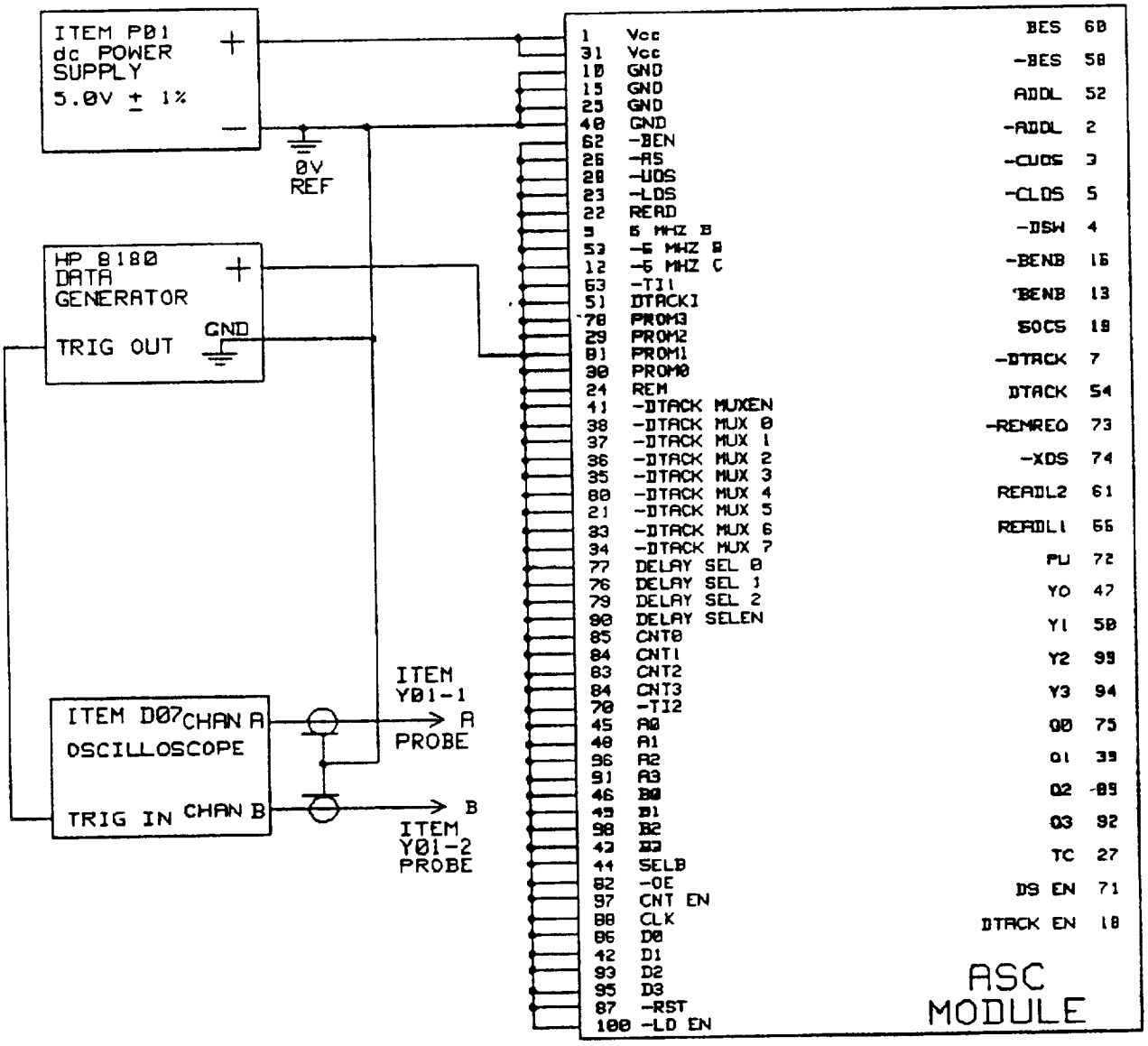


FIGURE 12. Output crosstalk, high and low level, test circuit for tables XX and XXI.

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TABLE XXII. Output crosstalk duration test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
V1	Y0	47	0	50	nS	1-4	13
V2	Y1	50					
V3	Y2	99					
V4	Y3	94	0	50	nS	1-4	13

NOTES:

1. Output crosstalk duration test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	V1-V4
Power supply current limit	0.913 A	NA	V1-V4
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 V1-V4 V1-V4
Crosstalk duration	NA	± 2.5 nS	V1-V4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

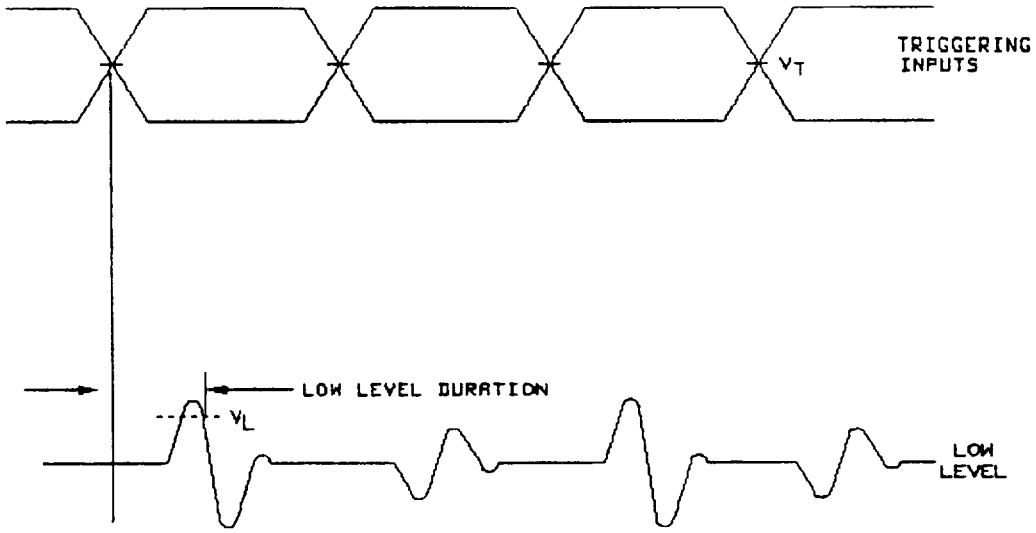
(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. Module inputs shall not violate the minimum input timing parameters referenced on figure 7.

(c) The crosstalk duration shall be measured from the triggering edge of the input as shown on figure 13.

3. All outputs shall be loaded as specified on figure 6.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. VT = 1.5 V, VL = 400 mV (ac coupled).

2. The crosstalk duration is measured from the triggering edge of the input to the point where the output crosstalk returns to the specified voltage. If the crosstalk amplitude never exceeds VL, the crosstalk duration shall be recorded as 0 nS.

FIGURE 13. Output crosstalk duration test waveforms for table XXII.

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TABLE XXIII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W1	A0	45	2	20	pF	1-3
W2	A1	48				
W3	A2	96				
W4	A3	91				
W5	B0	46				
W6	B1	49				
W7	B2	98				
W8	B3	43				
W9	D1	42				
W10	D2	93				
W11	D3	95				
W12	-RST	87				
W13	-AS	26				
W14	REM	24				
W15	-BEN	62				
W16	-DTACK MUX 6	33				
W17	-LD EN	100				
W18	CLK	88				
W19	SELB	44	2	20	pF	1-3
W20	-OE	82	2	25	pF	1-3
W21	D0	86	2	25	pF	1-3

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TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
W22	-DTACK MUX 0	38	2	15	pF	1-3
W23	-DTACK MUX 1	37				
W24	-DTACK MUX 2	36				
W25	-DTACK MUX 3	35				
W26	-DTACK MUX 4	80				
W27	-DTACK MUX 7	34				
W28	-DTACK MUXEN	41	2	15	pF	1-3
W29	CNT EN	97	4	30	pF	1-3
W30	READ	22	4	30	pF	1-3
W31	DELAY SEL 0	77	2	40	pF	1-3
W32	DELAY SEL 2	79				
W33	CNT1	84				
W34	CNT2	83				
W35	CNT3	32				
W36	-DTACK MUX 5	21	2	40	pF	1-3
W37	PROM0	30	4	40	pF	1-3
W38	-UDS	28				
W39	-LDS	23	4	40	pF	1-3
W40	DELAY SEL 1	76	2	50	pF	1-3
W41	DELAY SELEN	90				
W42	CNT0	85	2	50	pF	1-3

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TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W43	PROM3	78	2	50	pF	1-3
W44	6 MHZ B	9	4	60	pF	1-3
W45	DTACKI	51	4	45	pF	1-3
W46	-6 MHZ C	12	4	45	pF	1-3
W47	PROM2	29	6	45	pF	1-3
W48	PROM1	81	6	45	pF	1-3
W49	-6 MHZ B	53	8	75	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	W1-W49
Capacitance meter voltage	15 mV	± 1.5 mV	W1-W49
Capacitance measurement	NA	± 0.9 pF	W1-W19
	NA	± 1.15 pF	W20,W21
	NA	± 0.65 pF	W22-W28
	NA	± 1.3 pF	W29,W30
	NA	± 1.9 pF	W31-W36
	NA	± 1.8 pF	W37-W39
	NA	± 2.4 pF	W40-W43

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA NA NA NA	± 2.8 pF ± 2.05 pF ± 1.95 pF ± 3.35 pF	W44 W45,W46 W47,W48 W49

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIV. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
X1	Y0	47	2	25	pF	1-3
X2	Y1	50				
X3	Y2	99				
X4	Y3	94	2	25	pF	1-3

NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	X1-X4
Capacitance meter voltage	15 mV	± 1.5 mV	X1-X4
Capacitance measurement	NA	± 1.15 pF	X1-X4

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIV.

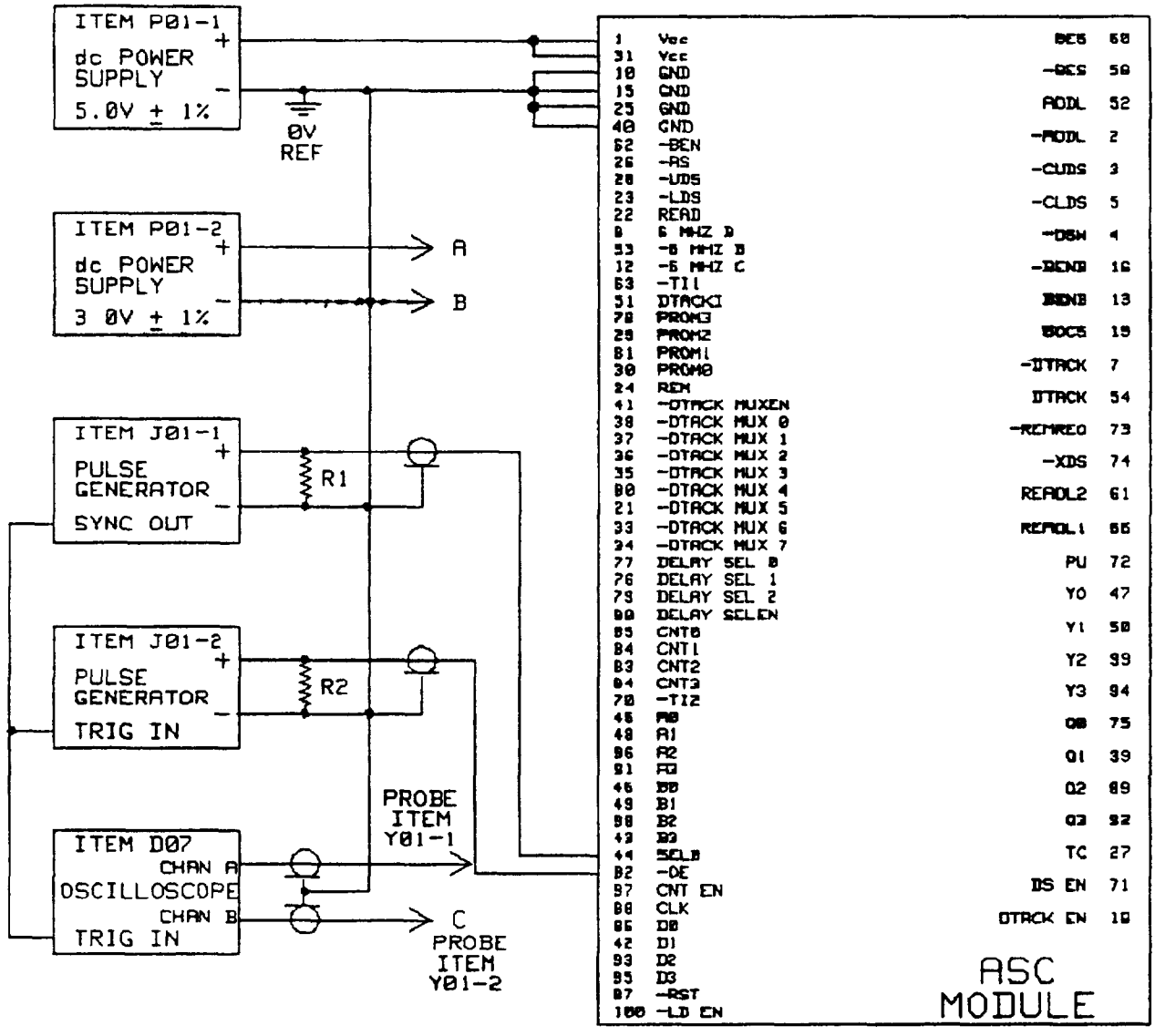
3. Abbreviations used are defined in 6.1.

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TABLE XXV. Disable time test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Low to high impedance (tpLz)								
Y1	-OE to Y0	45,48,96,91	46,49,98,43	47	3	31	nS	14,15
Y2	-OE to Y1			50				
Y3	-OE to Y2			99				
Y4	-OE to Y3			94				
High to high impedance (tpHz)								
Y5	-OE to Y0			47	3	69	nS	14,15
Y6	-OE to Y1			50				
Y7	-OE to Y2			99				
Y8	-OE to Y3	45,48,96,91	46,49,98,43	94	3	69	nS	14,15

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NOTES:

1. Output pins 47, 50, 99, and 94 shall be loaded as specified on figure 6. The load power supply shall be set at 2.1 V ± 1 percent.

2. R1 = R2 = 51 Ohms ± 5 percent, 1/2 W.

FIGURE 14. Disable time test circuit for table XXV.

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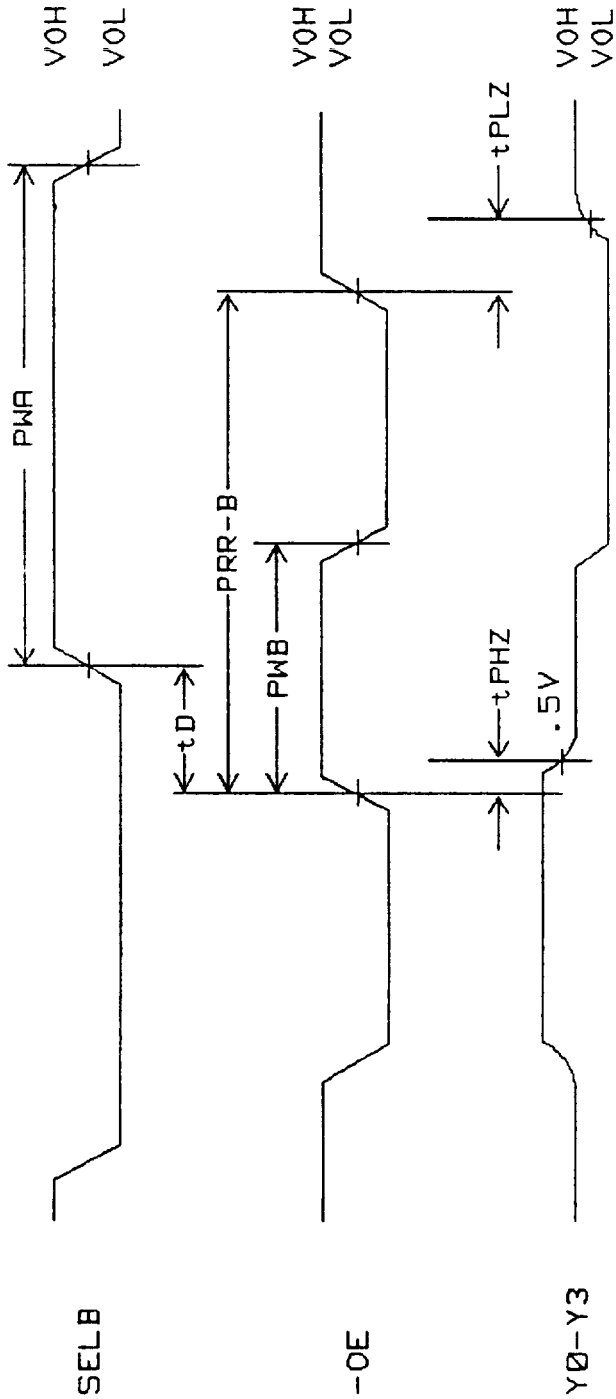


FIGURE 15. Disable time test waveforms for table XXV.

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NOTES:

1. PWA = 500 ± 50 nS.
2. PWB = 250 ± 25 nS.
3. PRR-B = 500 ± 50 nS.
4. tD = 125 ± 15 nS.
5. Transition times shall not exceed 15 nS.

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TABLE XXVI. Threshold test.

Test cond	Parameter	Notes	Fig
Z	Threshold	1-4	7

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Z
Power supply current limit	0.913 A	NA	Z
Pattern voltage			
high, (VH)	See note 3		Z
low, (VL)	See note 3		Z
Output voltage measurement			
high, (Voh)	NA	± 125 mV	Z
low, (Vol)	NA	± 190 mV	Z
low, (Vol)	NA	± 185 mV	Z

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results to the output low and high levels specified in the output voltage tests. The module inputs shall not violate the minimum input timing parameters referenced on figure 7.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. The input threshold levels are as follows:

Input pin	(VH)	(VL)
45,48,96,91,46,49,98,43,82,44, 86,42,93,95,88,97,87,100,38, 37,36,35,80,21,33,34,41,30,81, 29,26,22,28,23	2.0 V ± 20mV	0.7 V ± 7mV
85,84,83,32,77,76,79,90,78,62, 51,9,53,12,24	2.0 V ± 20mV	0.8 V ± 8mV

4. Abbreviations used are defined in 6.1.

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APPENDIX

MODULE FUNCTION

10. SCOPE

10.1 Scope. This appendix contains information regarding the module function. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

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APPENDIX

30. Module function. The ASC module provides dedicated support functions to a 68000 based microprocessor being operated at up to 6 MHz. The ASC module contains DTACK and data strobe support logic. A PAL is used to generate complex signals DS EN and DTACK EN. The boolean equations for these outputs are on figure 1.

$$\begin{aligned}
\text{DS EN} = {} & \overline{\text{PROM3}} \cdot \overline{\text{PROM2}} + \overline{\text{PROM3}} \cdot \text{DELAY SELEN} \cdot \text{BES} + \text{PROM3} \cdot \overline{\text{PROM1}} \cdot \text{DELAY SELEN} \cdot \text{BES} \\
& + \text{PROM3} \cdot \text{PROM1} \cdot (\text{DELAY SEL2} \cdot \text{DELAY SEL1} \cdot \text{DELAY SEL0} + \overline{\text{DELAY SEL2}} \cdot \overline{\text{DELAY SEL1}} \cdot \overline{\text{DELAY SEL0}}) \\
& + \text{PROM3} \cdot \text{PROM1} \cdot \text{DELAY SELEN} \cdot (\text{BES} \cdot \overline{\text{DELAY SEL2}} + \text{BES} \cdot \text{DELAY SEL2} \cdot \overline{\text{DELAY SEL1}} \cdot \overline{\text{DELAY SEL0}} + \text{CNT1} \cdot \text{DELAY SEL2} \cdot \text{DELAY SEL1} \cdot \text{DELAY SEL0}) \\
\text{DTACK EN} = {} & \overline{\text{PROM3}} \cdot \overline{\text{-XDS}} \cdot \text{BES} + \text{PROM3} \cdot \text{PROM1} \cdot (\overline{\text{DELAY SEL2}} \cdot \overline{\text{DELAY SEL1}} \cdot \overline{\text{DELAY SEL0}} \\
& + \text{DELAY SEL2} \cdot \text{DELAY SEL1} \cdot \text{DELAY SEL0}) + \text{PROM3} \cdot \overline{\text{PROM1}} \cdot \text{CNT1} \cdot \text{CNT0} \\
& + \text{PROM3} \cdot \text{PROM1} \cdot (\overline{\text{DELAY SEL2}} \cdot \overline{\text{DELAY SEL1}} \cdot \text{DELAY SEL0} \cdot \text{CNT1} + \text{DELAY SEL2} \cdot \overline{\text{DELAY SEL1}} \cdot \text{CNT2} \cdot \text{CNT0} + \overline{\text{DELAY SEL2}} \cdot \text{DELAY SEL1} \cdot \overline{\text{DELAY SEL0}} \cdot \text{CNT2} \\
& + \overline{\text{DELAY SEL2}} \cdot \text{DELAY SEL1} \cdot \text{DELAY SEL0} \cdot \text{CNT3} \cdot \text{CNT2} \cdot \text{CNT0})
\end{aligned}$$

FIGURE 1. ASC PAL boolean equations.

An "external DTACK" multiplexer is used to select the DTACK input pin. When a binary code is placed on inputs PROM0-PROM2 (LSD to MSB), the corresponding DTACK input (-DTACK MUX 0 - -DTACK MUX 7) is selected and synchronized to the rising edge of 6 MHz B. The synchronized output (-XDS) becomes active when the selected input goes low. PROM0-PROM3 are encoded inputs that give the module information about the peripheral being accessed. The state of PROM3, for example, tells the ASC module whether the DTACK signal will be generated by the module being accessed, or generated locally on the ASC module. When PROM3 is low, the DTACK signal must be generated externally, and PROM0-PROM3 select the input pin for the awaited DTACK signal (see table I). When PROM3 is high, the DTACK signal is generated locally by decoding other PAL inputs. The number of wait states to be inserted before a locally generated DTACk signal is determined by the DELAY SEL inputs (DELAY SEL 0-DELAY SEL 2), the CNT inputs (CNT0-CNT3), and PROM1. Table I shows the number of wait states for various input combinations when the general purpose counter outputs (Q0-Q3) are connected to CNT0-CNT3, respectively, and the counter is running at the same clock frequency as the 68000. The number of wait states can be altered by changing the clock frequency of the counter, or by preloading the counter with a number other than 0.

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TABLE I. Truth table of DTACK EN logic.

PROM inputs				DELAY SEL inputs			DTACK source	Number of wait states*
3	2	1	0	2	1	0		
0	0	0	0	X	X	X	-DTACK MUX 0 Input	Determined by DTACK source
0	0	0	1	X	X	X	-DTACK MUX 1 Input	Determined by DTACK source
0	0	1	0	X	X	X	-DTACK MUX 2 Input	Determined by DTACK source
0	0	1	1	X	X	X	-DTACK MUX 3 Input	Determined by DTACK source
0	1	0	0	X	X	X	-DTACK MUX 4 Input	Determined by DTACK source
0	1	0	1	X	X	X	-DTACK MUX 5 Input	Determined by DTACK source
0	1	1	0	X	X	X	-DTACK MUX 6 Input	Determined by DTACK source
0	1	1	1	X	X	X	-DTACK MUX 7 Input	Determined by DTACK source
1	X	0	X	X	X	X	Internal ASC Generated	3
1	X	1	X	0	0	0	Internal ASC Generated	0
1	X	1	X	0	0	1	Internal ASC Generated	2
1	X	1	X	0	1	0	Internal ASC Generated	4
1	X	1	X	0	1	1	Internal ASC Generated	13
1	X	1	X	1	0	0	Internal ASC Generated	5
1	X	1	X	1	0	1	Internal ASC Generated	5
1	X	1	X	1	1	0	NOT USED	
1	X	1	X	1	1	1	Internal ASC Generated	0

* Number of wait states assumes the Q0-Q3 are connected to CNT0-CNT3, that the counter is operating at the same frequency as the 68000 MPU, and that the counter is preloaded with 0000.

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APPENDIX

The module contains a remote request latch and a READ/-WRITE latch. When PROM2 and PROM1 are high, the remote request output (-REMREQ) becomes active on the rising edge of the remote request input (REM). -REMREQ remains active until -AS or -UDS, -LDS, and READ go high. READL1 and READL2 are outputs of a transparent latch. When ADDL goes high, READL1 and READL2 will be latched low until ADDL goes low again. A quadruple 2 input multiplexer is fully pinned out for general purpose use. When SELB is low, inputs A0-A3 correspond to outputs Y0-Y3, respectively, and when SELB is high, B0-B3 correspond to Y0-Y3. -OE is the output enable signal. When -OE is low, Y0-Y3 are enabled. When -OE is high, Y0-Y3 are in tristate. A four bit synchronous binary counter is pinned out for general purpose use. Parallel loading is accomplished by placing data on D0-D3 and driving -LD EN low. On the next rising edge of the clock (CLK), the input data is preloaded into the counter. When CNT EN, -RST, and -LD EN are high, the binary outputs (Q0-Q3) synchronously increment on each rising edge of the clock. When all four Q outputs are high, TC will go high. CNT EN is a count enable input. When CNT EN is low, counting is stopped and TC is disabled. The asynchronous reset input (-RST) clears all outputs (Q0-Q3). The ASC module generates bus control strobes which support both 68000 microprocessor and DMA controlled data transfers. When -BEN is active, it indicates that the 68000 is controlling the address and data busses, and that input signals (-AS, -UDS, -LDS, and READ) are synchronous to the clock (6 MHZ B). When -BEN is inactive, it indicates that a DMA controller or other supervisor is performing a transfer. In such a case, -AS, -UDS, -LDS, and READ are assumed to be asynchronous. The SOCS signifies the start of a bus cycle when -AS, and -UDS, -LDS or READ go low. On the next rising edge of the clock (6 MHZ B), BES and -BES become active. The bus control strobe generation logic then waits until the DS EN inputs (PROM3, PROM2, BES, DELAY SELEN, DELAY SEL 0, DELAY SEL 1, and DELAY SEL 2) meet the conditions required to generate DS EN. Once DS EN becomes active, the next rising edge of the clock enables the conditioned data strobes. -DSW goes low if READ is low, -CUDS goes low if -UDS is low, and -CLDS goes low if -LDS is low. At this point, the circuit waits for DTACK EN to be generated. On the rising edge of the clock after DTACK EN is generated, DTACK and -DTACK become active. On the rising edge of the clock following DTACK generation, ADDL and -ADDL become active. When -AS or -UDS, -LDS and READ are negated by the bus master, SOCS, DTACK, -DTACK, -DSW, -CUDS and -CLDS are returned to their inactive states. If ADDL has been active for at least one clock period before the bus master signals were

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negated, then BES and -BES are also negated. If the bus master strobes are negated before ADDL has been active for at least one clock period, then BES and -BES remain active until the next rising edge of the clock. ADDL and -ADDL are negated after the rising edge of the clock following BES negation. Figure 2 shows the timing of a typical ASC bus cycle.

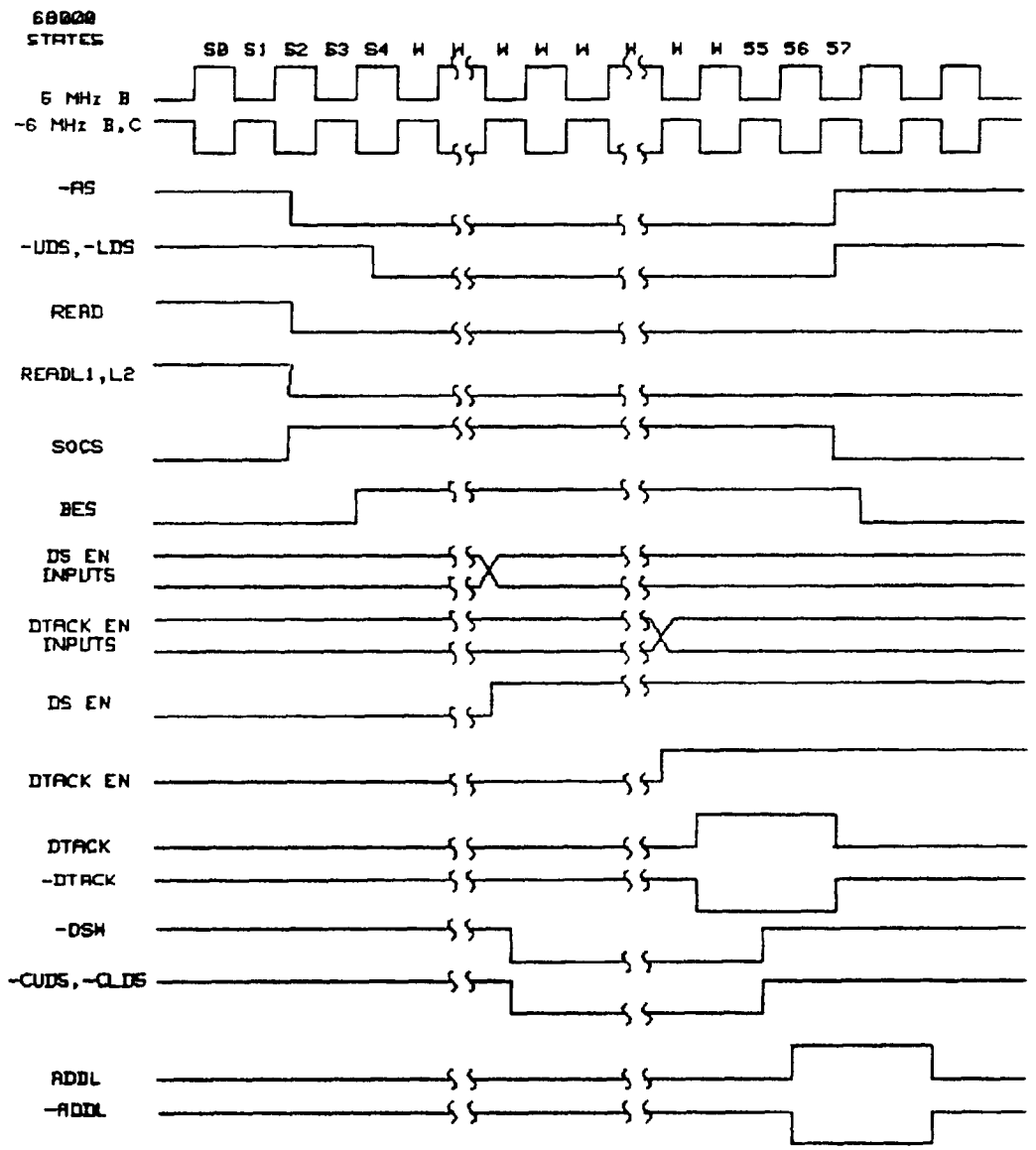


FIGURE 2. Typical ASC timing for a 68000 write cycle.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	MIL-W-28787/351	1990 October 26

3 DOCUMENT TITLE

Modules, Digital, Standard Electronic, 68000 Support Module, Key Code ASC

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code)	
Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A)	(1) Commercial	(2) AUTOVON
PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 812-854-1528	88-482-1528
ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Robert F. Dilger, Bldg. 2940W, Code 602A Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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